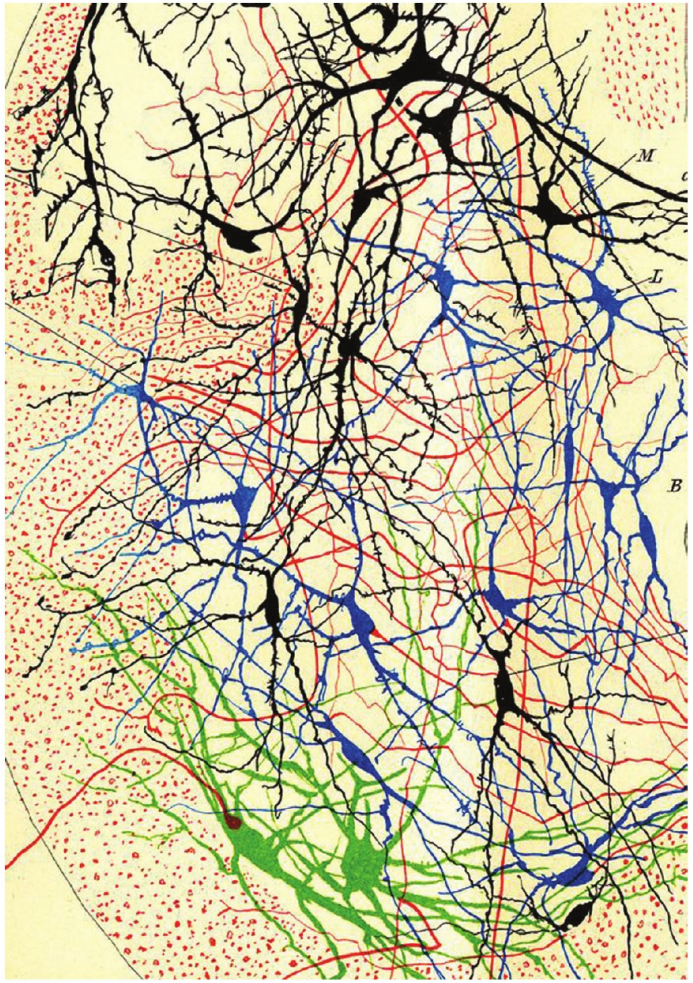


Science Source

Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal’s drawing of Golgi-stained neurons.

BEHAVIORAL NEUROSCIENCE 392

Foundations and Frontiers

From Brain to Behavior: Foundations and Frontiers of Behavioral Neuroscience

Fall 2026

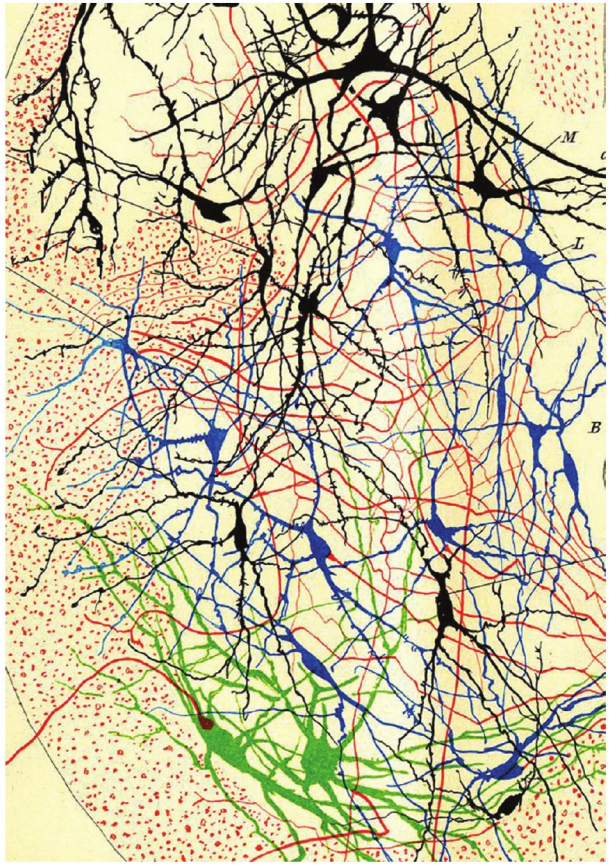
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Behavior links brain activity with the environment

Behavioral neuroscience studies how biological processes produce perception, movement, learning, emotion, and other behavior. A **neuron** is a nerve cell that communicates with other cells. Groups of connected neurons form circuits whose activity changes as an organism interacts with its environment.

The nervous system receives information from the senses and uses that information to organize actions. Perception helps an animal locate food, avoid danger, and communicate. Memory allows previous experience to influence the response to a new situation.

Psychology contributes methods for describing behavior, while physiology investigates how organs and cells work. Behavioral neuroscience combines these approaches in the same explanation. A learning experiment can measure an animal's choices and investigate the neural activity that accompanies those choices.



Science Source

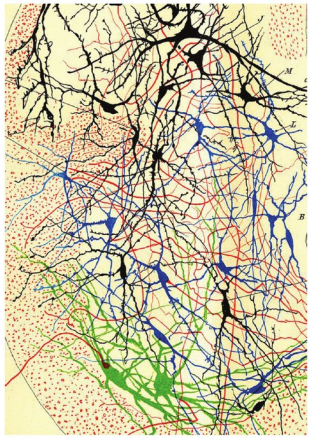
Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.

Plasticity changes connections across a lifetime

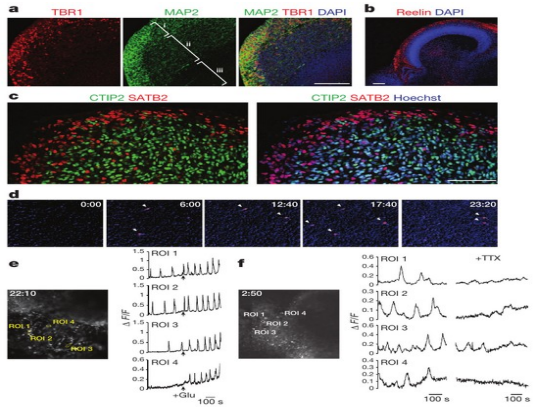
Plasticity is the capacity of neural systems to change through growth, altered connections, and reorganization. Learning and experience can change relationships among neurons. The resulting network can respond differently when a familiar situation occurs again.

The textbook opens with a child affected by severe seizures associated with Rasmussen encephalitis. A **seizure** involves abnormal neural activity that can disrupt voluntary behavior. An inflamed cerebral hemisphere can become the source of repeated seizures.

A **hemisphere** is one of the two large sides of the cerebrum. After major childhood surgery, rehabilitation provides repeated opportunities to practice affected skills. The textbook relates recovery to developing networks that reorganize their connections and functions.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



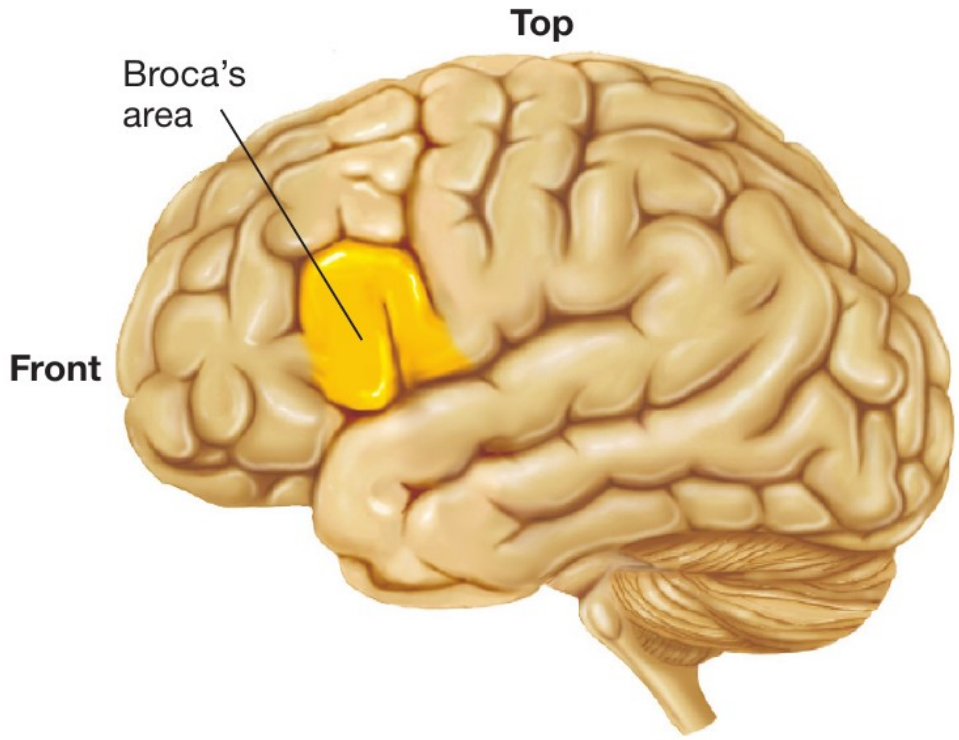
Visual context: Lancaster et al. (2013), Supplementary Video 2. Calcium surges.

Neural organization supports coordinated functions

Neurons operate in connected systems rather than as isolated cells. Sensory information can influence movement through connections linking multiple brain regions. Communication between regions allows perception, remembered information, and the current action to affect one another.

The cerebral cortex is the outer sheet of the cerebrum. Different cortical regions contribute to functions such as vision, movement, and language. **Broca's area** is a region in the left frontal part of the brain historically associated with speech production.

Damage to a region can change the behavior that depends on it. Such observations connect a change in behavior with the organization of the nervous system. Anatomical location, communication among regions, and experience-dependent reorganization all contribute to an account of function.



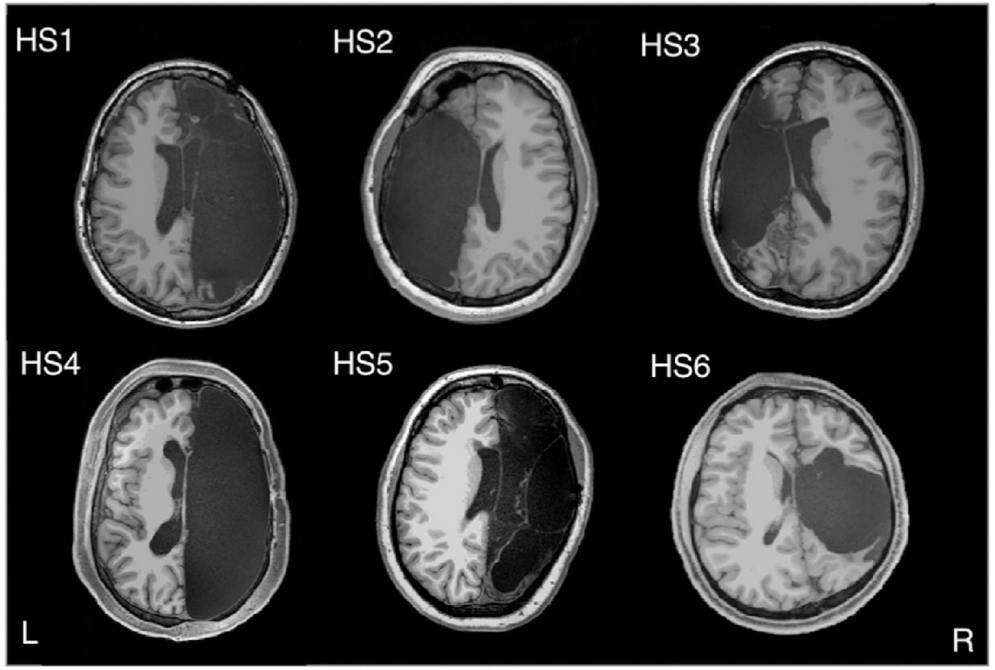
Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca's area in the left hemisphere.

A single hemisphere retains organized brain networks

Kliemann and colleagues studied adults who had undergone **hemispherectomy**, surgical removal or disconnection of a cerebral hemisphere, during childhood. Structural magnetic resonance imaging, or **MRI**, documented the remaining brain anatomy. The participants had experienced extensive loss of one side of the cerebrum early in development.

The study also measured activity while participants rested in the scanner. A **brain network** is a group of regions whose activity is organized together. Regions within the surviving hemisphere remained organized into recognizable networks associated with vision, movement, attention, and other functions.

The remaining tissue retained relationships among regions that are also observed in people with two cerebral hemispheres. The researchers examined how those relationships were arranged within and across networks. These measurements connect childhood changes in anatomy with the organization of adult brain activity.



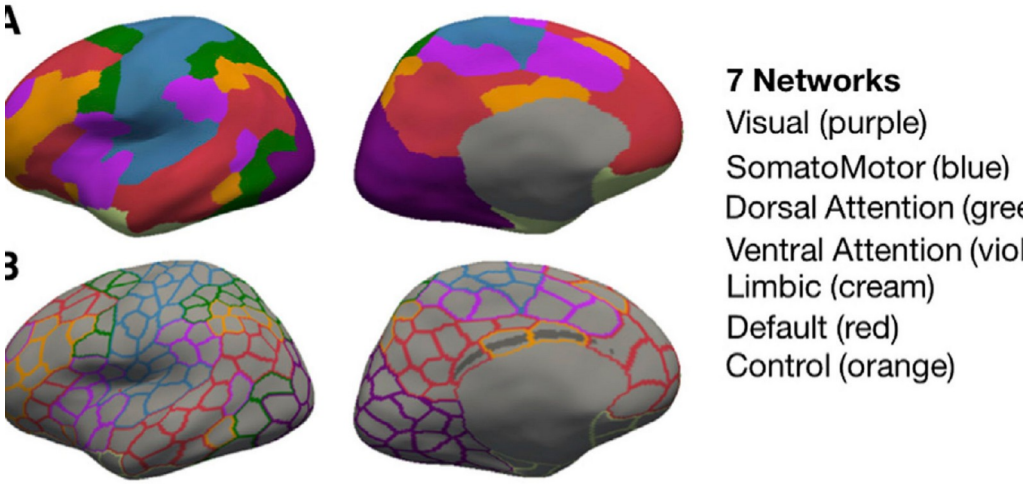
Kliemann et al. (2019), Fig. 1. Anatomy after childhood hemispherectomy.

Functional connectivity measures coordinated fluctuations

Functional MRI, or **fMRI**, follows changes in signals associated with blood oxygenation. The **blood-oxygenation-level-dependent** signal is commonly abbreviated **BOLD**. Neural activity and local blood supply are related, allowing researchers to measure slow changes associated with regional brain function.

Functional connectivity describes the relationship between the changing activity signals of different regions. Kliemann and colleagues compared these signals across cortical areas during rest. Regions assigned to the same network tended to fluctuate together.

The cortical map separated visual, movement-related, attention-related, and other networks. A map of regions provided locations for comparing the activity signals. The analysis connected a spatial description of the cortex with the way its signals changed over time.



Kliemann et al. (2019), Fig. 2A–B. Networks and regional parcellation.

Brain networks couple more strongly after surgery

The study distinguished relationships within a network from relationships between different networks. Participants with childhood hemispherectomy retained strong within-network organization. Their remaining hemisphere contained coordinated systems rather than a uniformly connected mass of tissue.

Relationships between different networks were stronger in almost all hemispherectomy participants. This pattern occurred across multiple network types. Regions belonging to different functional systems therefore had more closely related activity fluctuations.

The authors proposed the **hypothesis** that increased interactions between networks characterize functional reorganization after childhood hemispherectomy. The surviving tissue combines recognizable functional systems with altered relationships among those systems. The findings describe a pattern of organization through which researchers can investigate developmental plasticity.

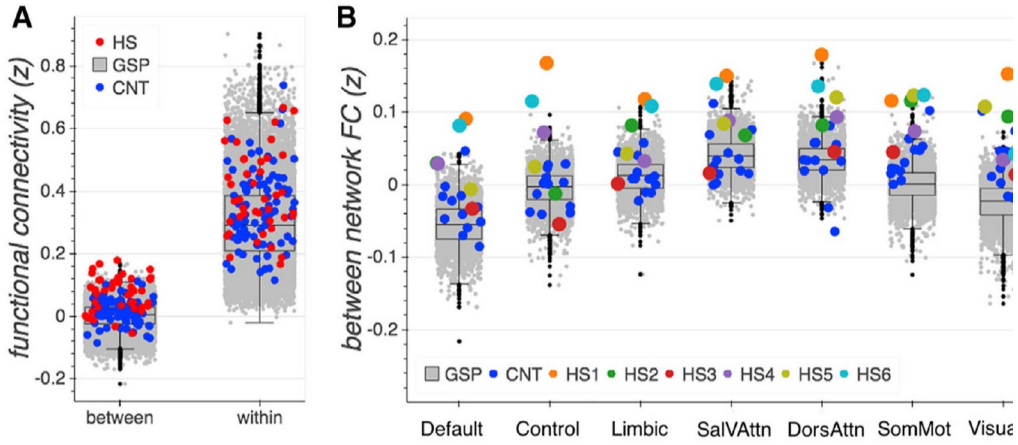


Figure 4. Functional Connectivity
(A) Between- and within-network functional connectivity averaged across networks per group (GSP, CNT, and HS) (seven data points, one for each network, per participant). CNT and HS showed similar within-network connectivity as compared to the large GSP sample, while overall between-network connectivity was notably stronger for HS participants. (B) High between-network connectivity was evident across all networks and in all but one

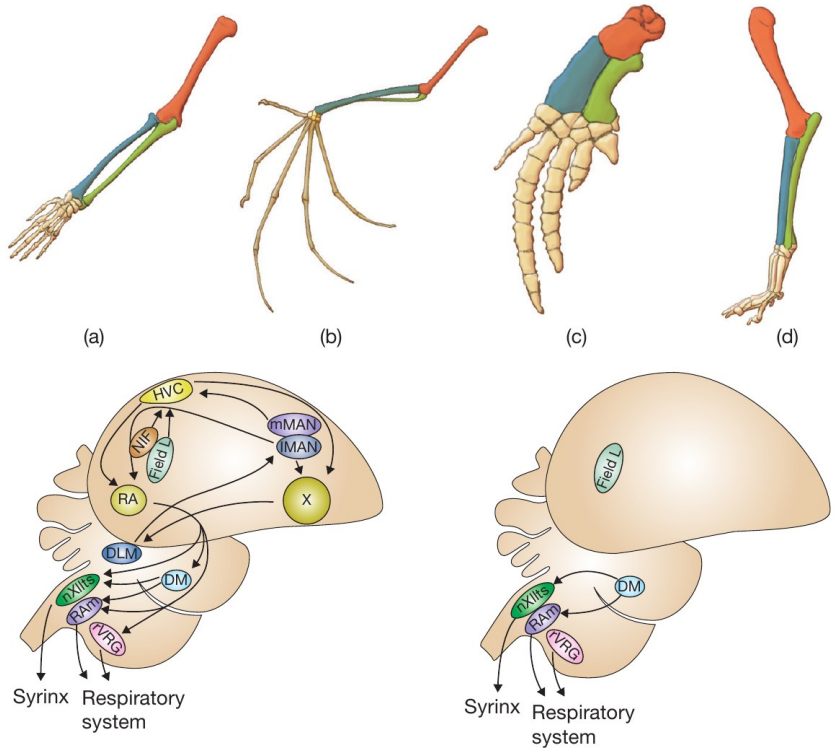
Kliemann et al. (2019), Fig. 4A–B. Within-network and between-network connectivity.

Generalization identifies recurring behavioral rules

Generalization explains an observation through a rule that applies across multiple examples. A researcher can compare behavior in several situations to identify a recurring relationship. The explanation describes what changes systematically when the conditions change.

Nest building depends on an animal's biological circumstances. A nonpregnant mouse may build a nest when the surroundings are cool. Pregnancy can also promote nest building, including conditions in which low temperature is not the immediate trigger.

The observable movements involved in constructing the nest can recur across these circumstances. Comparing circumstances identifies the situations in which the behavior occurs. The biological explanation also includes the different bodily signals that recruit the shared action.



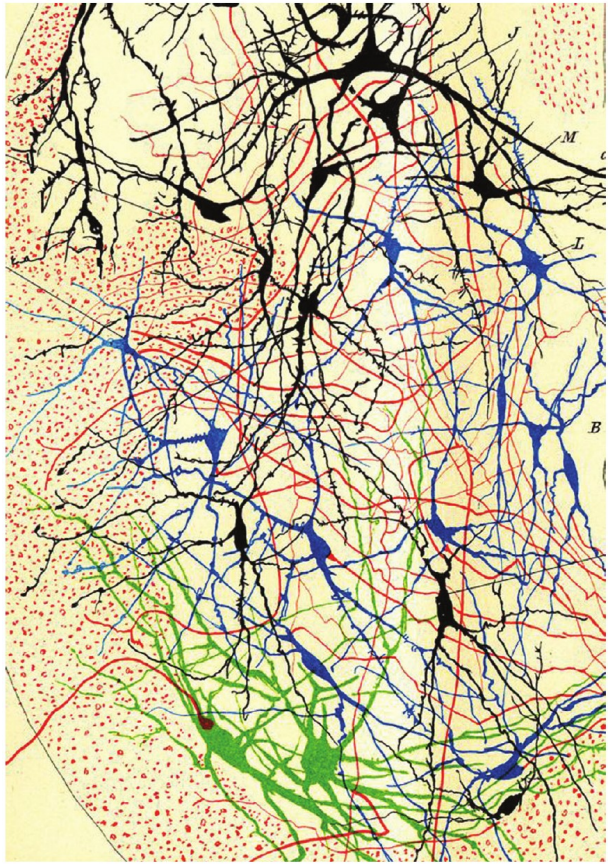
Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.

Reduction connects behavior with biological processes

Reduction explains a complex phenomenon in terms of its component processes. A behavioral neuroscientist can relate a pattern of movement to the activity of neural circuits. The investigation can then examine the signals that influence those circuits.

In the nest-building example, temperature-related signals and pregnancy-related hormones provide different sources of information. A **hormone** is a chemical signal released within the body that influences target cells. These signals can affect systems that organize nest-building movements.

Generalization describes relationships across behavioral examples, while reduction describes the processes that produce them. Both are part of a biological explanation. The same motor action can be recruited through different physiological routes according to the animal's condition.



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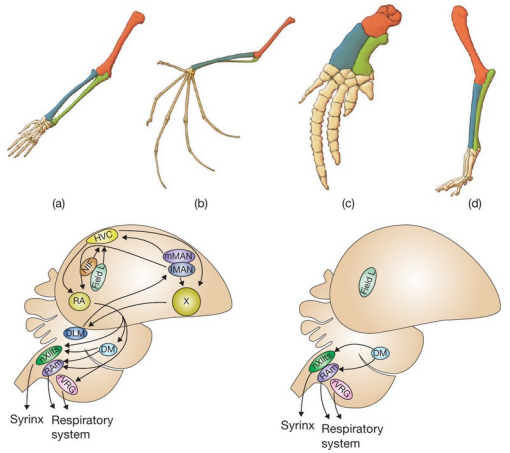
Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.

Behavioral function and neural mechanism are connected

A **mechanism** is the set of processes that produces an outcome. A behavioral explanation includes the signals entering a system, the operations within it, and the action that follows. The biological function concerns what that action accomplishes for the organism.

A nest provides an environment that can help maintain body temperature and support offspring. Nest construction requires a sequence of movements coordinated by the nervous system. Temperature and reproductive state can change the circumstances in which that sequence is recruited.

The mechanism and function refer to different aspects of the same behavior. A physiological signal can explain when construction begins, while the resulting nest changes conditions experienced by the animal. Perception, internal state, and movement jointly contribute to the outcome.



Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.



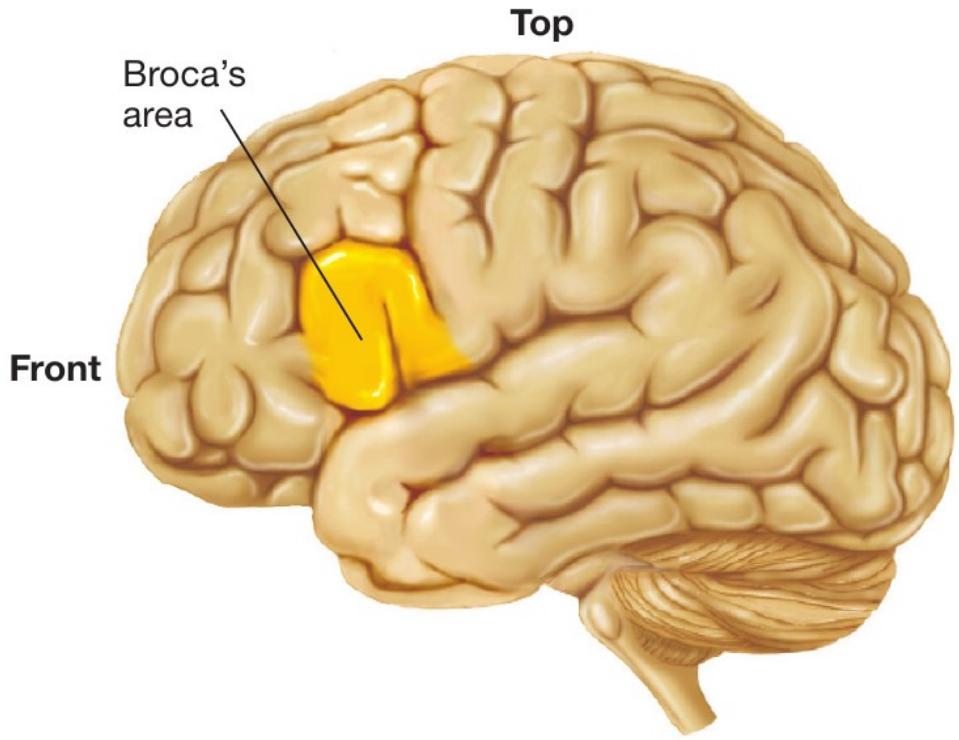
Visual context: MBARI, natural-history footage. Barreleye (Macropinna microstoma).

Biological explanations replaced early heart-centered ideas

Early explanations assigned mental functions to different organs. Hippocrates associated thought and emotion with the brain. Aristotle proposed a major role for the heart and described the brain as a cooling organ.

Galen investigated the nervous systems of several animal species through anatomical study. Sensory nerves connected sense organs with the brain. Such anatomical relationships supported a brain-centered account of sensory and behavioral functions.

The history reflects a shift toward explanations tied to observable biological structures. Connections among organs provided evidence that could be examined directly. Later investigations related damage to particular structures with changes in behavior.



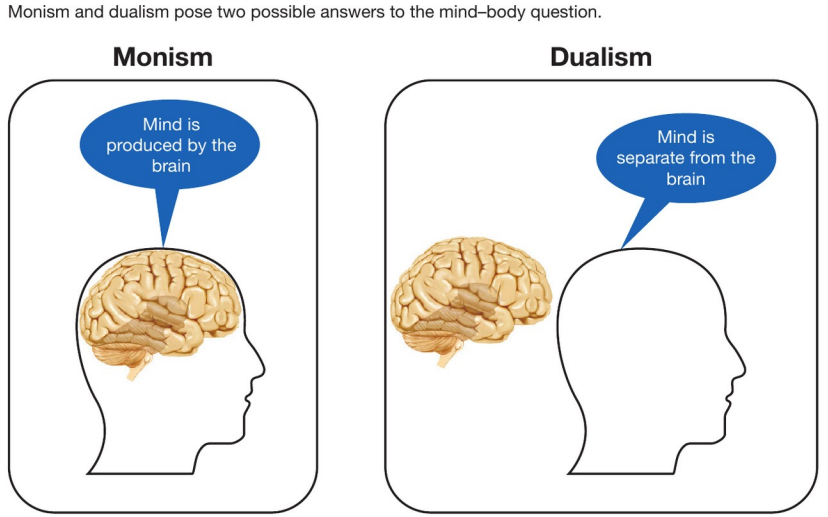
Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca's area in the left hemisphere.

Monism treats mental activity as brain function

Dualism treats mind and body as different kinds of entities. **Monism**, in the textbook's discussion of neuroscience, treats mental activity as a function of the physical brain. These positions supplied different answers to the historical mind–body question.

Descartes proposed that some behavior could arise mechanically through the nervous system. He also proposed an interaction between a nonphysical mind and the body through the pineal gland. The textbook presents this proposal as a historical explanation.

Behavioral neuroscience investigates the biological events associated with perception, memory, and action. It measures activity and examines how altering a biological process changes behavior. Mental functions are approached through relationships among physical systems and observable responses.



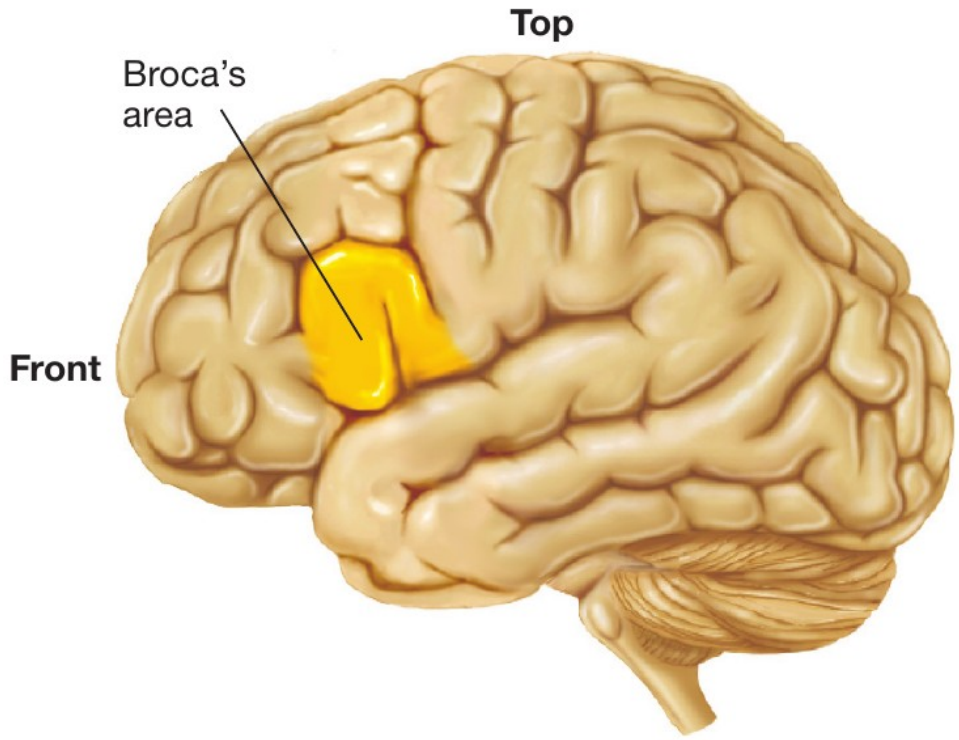
Carlson and Birkett (2022), Fig. 1.2, p. 19. Historical accounts of the mind–body relationship.

Anatomical observations localized speech-related function

Paul Broca studied a person whose brain injury was associated with a major loss of speech production. Examination identified damage in the left frontal region. This observation linked a behavioral change with a specific anatomical location.

Localization of function refers to the contribution of particular brain regions to particular functions. Language production includes processes that transform information into coordinated speech. Damage can disrupt part of that organization.

Broca’s work connected a damaged anatomical region with a disruption of speech. Clinical observations contributed to biological explanations of mental functions. Modern measurements extend that approach by examining both regional activity and relationships among regions.



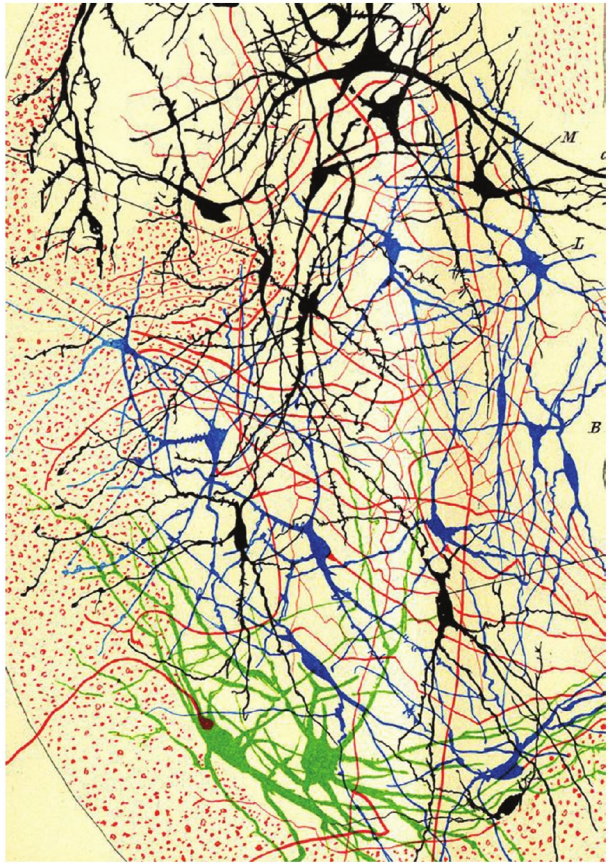
Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca’s area in the left hemisphere.

Staining revealed the cellular organization of the brain

A **stain** is a preparation used to make biological structures easier to distinguish. Golgi's staining method made individual nerve cells visible within nervous tissue. Their cell bodies and branching processes could then be examined under a microscope.

Ramón y Cajal studied stained cells and drew their detailed shapes. The branching structures revealed the organization of individual neurons within the tissue. His observations supported the view that the nervous system contains distinct cells.

Cellular anatomy connected a complex organ with its smaller structural components. The structure of a neuron can be studied together with its physiological activity and connections. This cellular approach became a foundation for research relating circuits to behavior.



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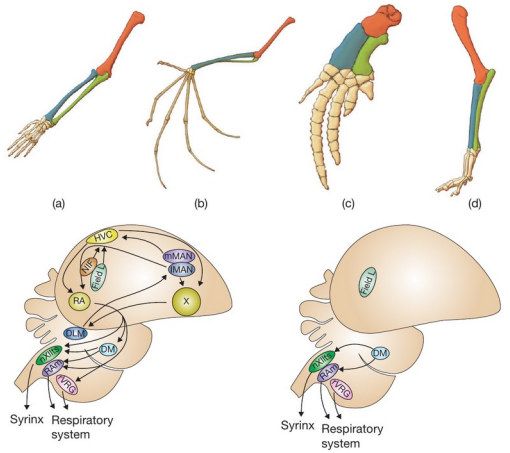
Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.

Selection changes inherited traits across generations

Natural selection occurs when inherited differences influence survival and reproduction. Individuals within a population vary, and some of that variation is transmitted to offspring. Traits associated with greater reproductive success can become more common across generations.

The environmental conditions affect which traits provide an advantage. A structure useful for obtaining one kind of food may be useful differently in another habitat. Selection concerns the interaction between an organism's characteristics and its circumstances.

Behavior can influence access to resources, avoidance of predators, and reproduction. The nervous system contributes to these actions through perception and movement. Biological explanations therefore include both how behavior is produced and how inherited characteristics change across generations.



Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.



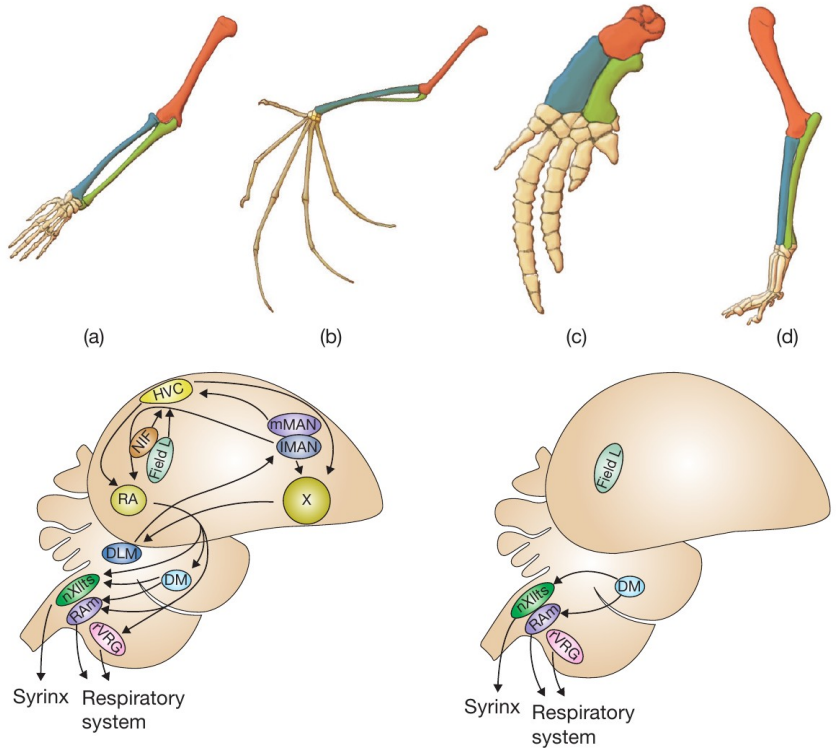
Visual context: MBARI, natural-history footage. Dragonfish (Idiacanthus antrostomus).

Functionalism relates traits to their biological roles

Functionalism interprets a characteristic in terms of the role it performs for an organism. A wing, a sensory organ, or a behavioral pattern can be examined in relation to survival and reproduction. Its usefulness depends on the environment and the organism's way of life.

Humans, bats, whales, and dogs have related forelimb structures. The underlying bones have been modified into structures used in different activities. A related anatomical plan can support manipulation, flight, swimming, or terrestrial movement.

The brain also contains structures whose organization is related to behavioral roles. Comparing species can connect differences in neural systems with differences in what animals do. An evolutionary explanation combines inherited structure, environmental conditions, and biological function.



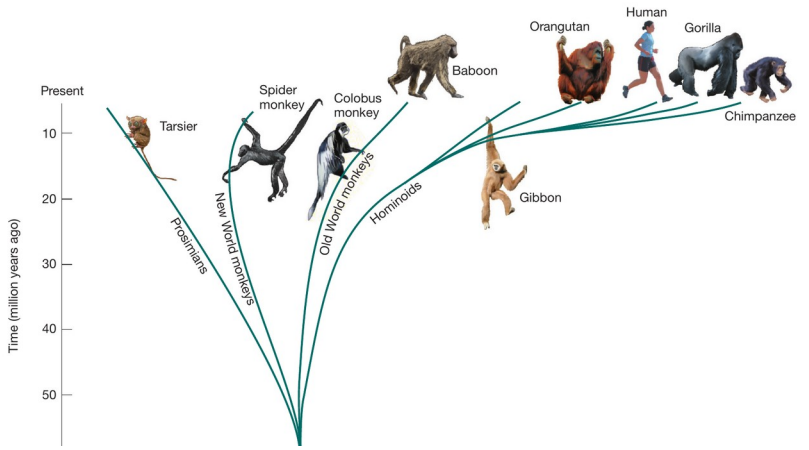
Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.

Mutation supplies new inherited variation

A **gene** is a segment of DNA involved in producing biological characteristics through the molecules it encodes. DNA is carried in chromosomes and transmitted during reproduction. Inherited differences contribute to variation among individuals.

A **mutation** is a change in genetic material. A mutation affecting reproductive cells can be passed to offspring. The consequences depend on how the change influences development and the organism's biological functions.

Natural selection acts on the differences that occur within a population. An inherited characteristic can become more frequent when it increases reproductive success under the prevailing conditions. Variation and selection together contribute to changes across generations.



Carlson and Birkett (2022), Fig. 1.8, p. 26. Evolutionary relationships among primates.



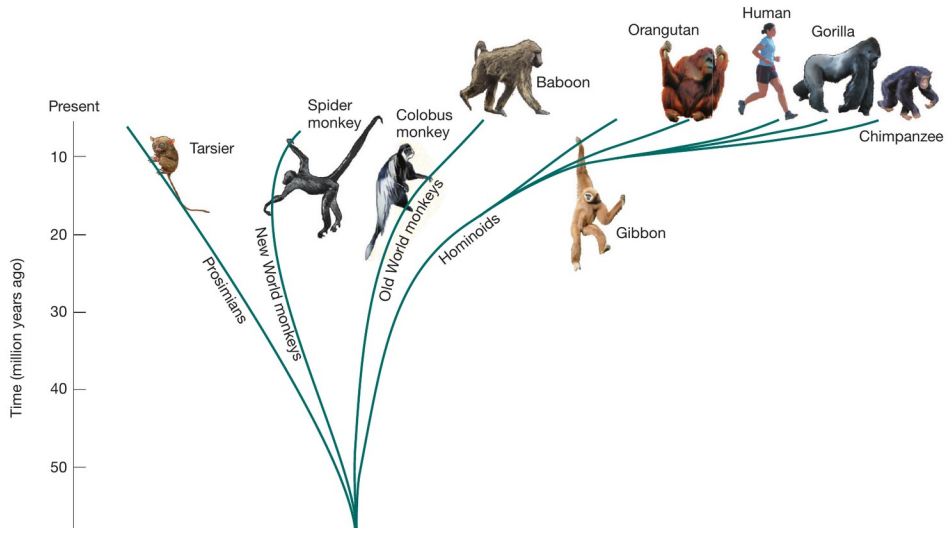
Visual context: MBARI, natural-history footage. Vampire squid (Vampyroteuthis infernalis).

Primate relationships connect brains with shared ancestry

Primates are the mammalian group that includes monkeys, apes, and humans. Species within this group share inherited characteristics while also differing in anatomy and behavior. A phylogenetic tree represents relationships through common ancestry.

A **phylogeny** describes the evolutionary relationships among organisms. The textbook's primate tree places humans among other living primates. Separate branches represent lineages that developed after ancestral populations diverged.

Comparisons among related species help organize questions about brain evolution. A shared feature can be investigated alongside a specialization that differs among lineages. Neural anatomy and behavioral abilities are examined within this history of inherited change.



Carlson and Birkett (2022), Fig. 1.8, p. 26. Evolutionary relationships among primates.

Brain mass and neuron number are different measurements

Brain mass measures the amount of brain tissue, while **neuron number** counts its nerve cells. Tissue also contains supporting cells and other components. Species can differ in the number of neurons contained within a given amount of tissue.

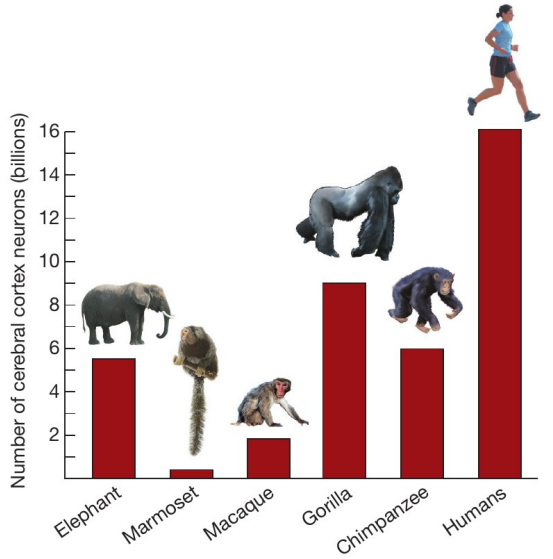
Researchers have compared the cellular composition of primate and rodent brains. Primate brains contain more neurons per gram in the comparisons discussed. A large brain and a densely populated brain therefore describe different anatomical properties.

The cerebral cortex contains circuits that contribute to perception, movement, and cognition. Comparing cortical neuron numbers provides information about how this tissue is organized across species. Brain evolution includes changes in size, cellular composition, and developmental processes.

Figure 1.10 Comparison of Mammalian Brains

Species with more complex behaviors have brains with more neurons that are available for behavior, learning, remembering, reasoning, and making plans. Primate brains contain more neurons per gram than rodent brains and more neurons in the cortex.

Source: Herculano-Houzel, S., and Marino, L. (1998.) A Comparison of Encephalization between Odontocete Cetaceans and Anthropoid Primates. *Brain, Behavior and Evolution, 51*(4), 230–238.



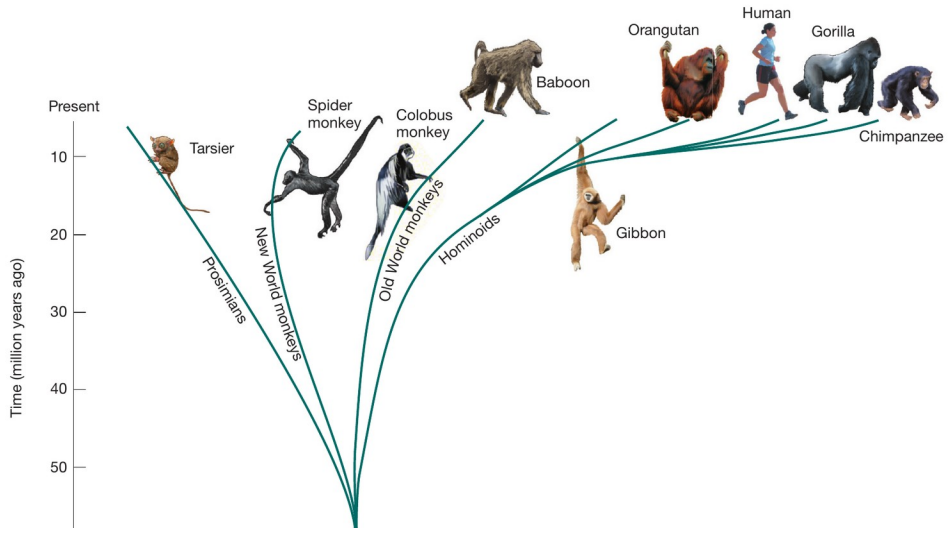
Carlson and Birkett (2022), Fig. 1.10, p. 27. Cortical neuron numbers across mammals.

Extended development contributes to human brain growth

Brain development includes the production of cells, growth of their processes, and formation of connections. The timing of these processes influences the final organization of the nervous system. Human brain growth continues extensively after birth.

Human brain evolution includes a prolonged developmental period. **Neoteny** refers to the retention or extension of juvenile developmental characteristics. A longer period of development provides more time for growth and experience-dependent change.

A growing brain develops within a social and physical environment. Experiences affect neural connections during that development. The organization of adult behavior reflects both inherited developmental processes and the information acquired during life.



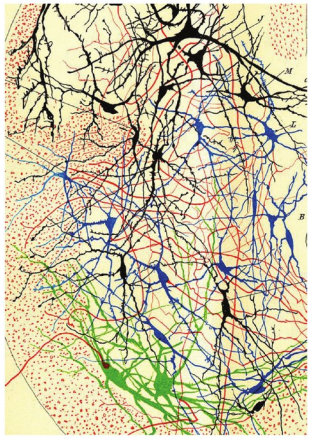
Carlson and Birkett (2022), Fig. 1.8, p. 26. Evolutionary relationships among primates.

Animal research includes care and ethical oversight

Animal research investigates biological functions in living nervous systems. Many cellular and physiological processes can be compared across species. Researchers also study how particular animals perceive, learn, and act within their environments.

The **Institutional Animal Care and Use Committee**, or **IACUC**, reviews animal-research procedures within an institution. Its work includes attention to animal care and proposed experimental methods. Laws and institutional requirements govern the conduct of this research.

Animal welfare concerns the conditions experienced by the animal and its biological needs. Housing, opportunities for appropriate behavior, and health care are part of research practice. Computerized enrichment can be investigated as an opportunity for voluntary engagement in managed animals.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



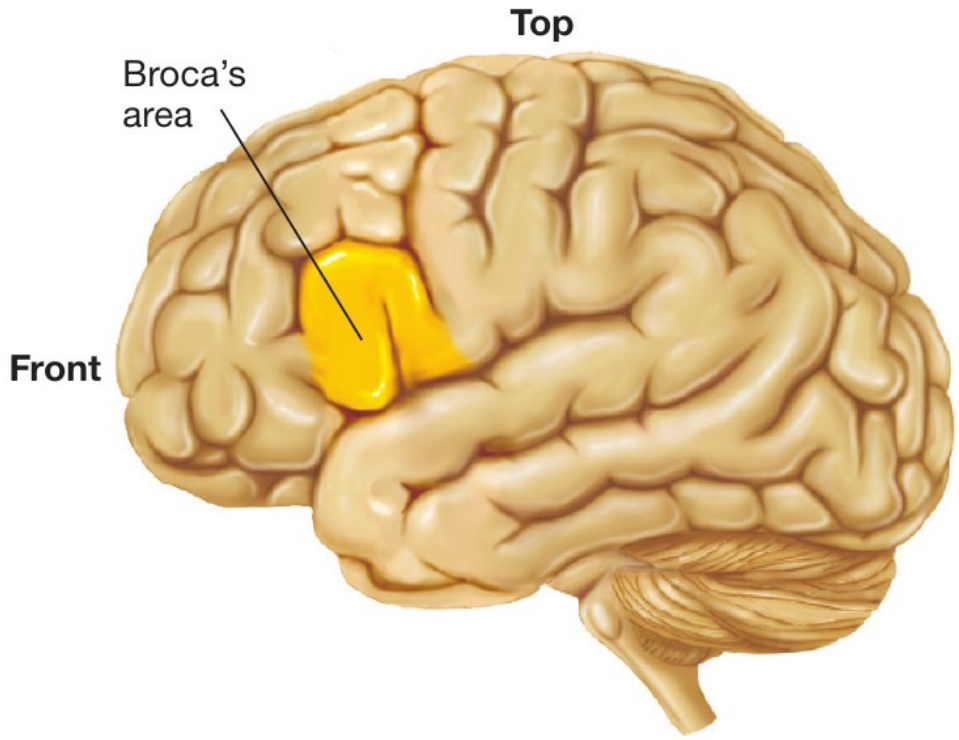
Visual context: Blackpool Zoo (2009). Bimbo balancing a ball.

Human research includes consent and privacy

Informed consent involves communicating relevant information so that a person can decide whether to participate in research. Human studies also require attention to privacy and protection of personal information. These responsibilities apply alongside the scientific aims.

The **Institutional Review Board**, or **IRB**, evaluates proposed research with human participants. The review considers the procedures and protections described in the research proposal. Institutional oversight brings scientific and participant-related considerations into the same evaluation.

Neuroethics examines ethical questions arising from research and applications involving the nervous system. Brain privacy, changes in mental functions, and the use of new technologies are included in this field. Neural interfaces connect measurements of brain activity with decisions about how the resulting information will be used.



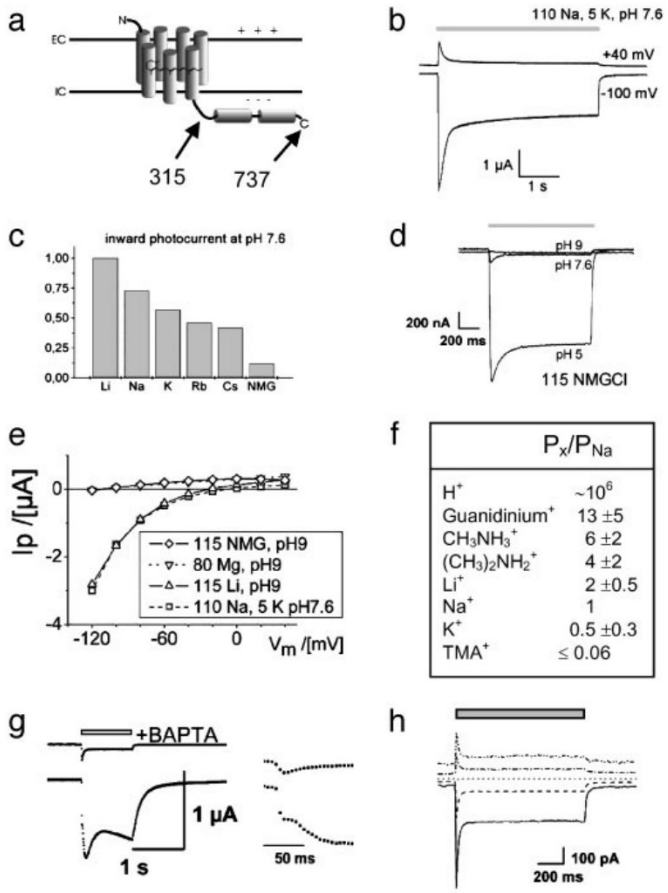
Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca's area in the left hemisphere.

A microbial channel directly couples light to current

Channelrhodopsin-2, or ChR2, is a membrane protein from the green alga *Chlamydomonas reinhardtii*. It contains **retinal**, a light-absorbing molecule. An **ion channel** provides a pathway for charged particles to move across the membrane. Nagel and colleagues expressed ChR2 in frog **oocytes**, immature egg cells, and mammalian cells to investigate its action.

Blue illumination near 450 nm produced currents absent from uninjected oocytes. Under **voltage clamp**, an amplifier holds the electrical potential difference across the cell membrane at a selected value while measuring current. Changing the held voltage changed current magnitude and direction, supporting a passive conductance rather than an ion pump.

A truncated version containing the first 315 amino acids retained channel function. Amino acids are the building blocks of proteins. The light-gated conductance remained within the membrane-spanning part of ChR2, linking illumination directly to an ion pathway across the cell membrane. Karl Deisseroth, Peter Hegemann, and Georg Nagel received the 2026 Nobel Prize in Physiology or Medicine for optogenetics.



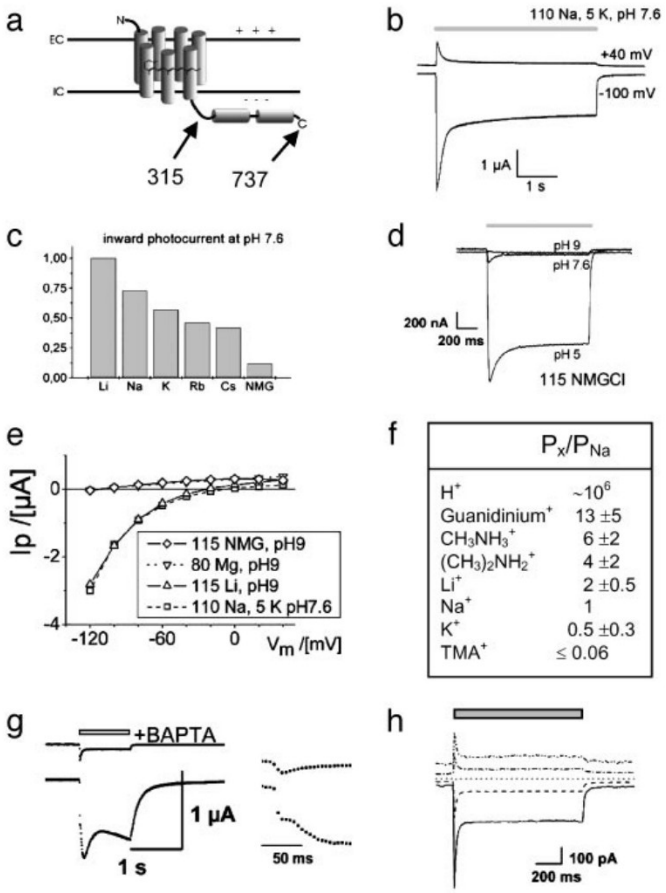
Nagel et al. (2003), Fig. 1(a–h). Original ChR2 currents under ion substitution and voltage clamp.

Different ions carry the light-evoked current

Ion selectivity describes which charged species can pass through a channel. Nagel and colleagues replaced salts around ChR2-expressing oocytes while measuring the response to the same blue-light stimulation. **Cations** carry positive charge; **anions** carry negative charge.

Photocurrents varied with the available cation. Replacing external chloride with aspartate did not appreciably change the current or its reversal voltage. **Reversal voltage** is the membrane voltage at which the net measured current changes direction. Calcium could also pass through ChR2, whereas magnesium permeability was negligible under the tested conditions.

At pH 9, the small current remaining in a solution containing the bulky cation N-methyl-D-glucamine disappeared. A **proton** is a positively charged hydrogen ion, and **pH** describes acidity. The result identified a proton contribution at lower pH, connecting the composition of the surrounding solution with the ions carrying the light-evoked current.



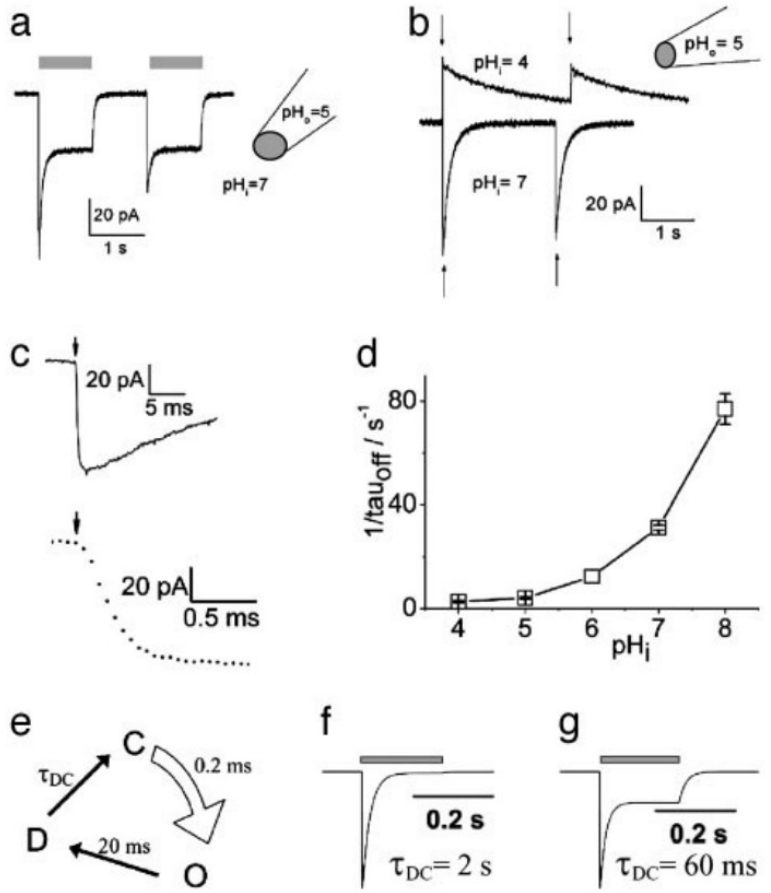
Nagel et al. (2003), Fig. 1(a–h). Currents and selectivity under controlled salt solutions.

Light gates the channel within the cell membrane

A light-evoked current could arise either from the expressed protein itself or from a signaling pathway that activates another channel. Nagel and colleagues recorded ChR2 in **excised membrane patches**, membrane fragments separated from the intact cell with their internal solution experimentally controlled.

Photocurrents remained after soluble cellular contents were replaced. A 10-ns laser flash triggered channel opening with a rise time near 200 µs or faster. Closing was much slower and depended on intracellular pH, spanning approximately 10–400 ms across the tested conditions.

The channel continued to respond rapidly in an excised membrane patch. The light-sensitive protein and its membrane environment therefore retained the machinery needed for gating. Different opening and closing times gave the electrical response a time course that extended across the illumination period and its termination.



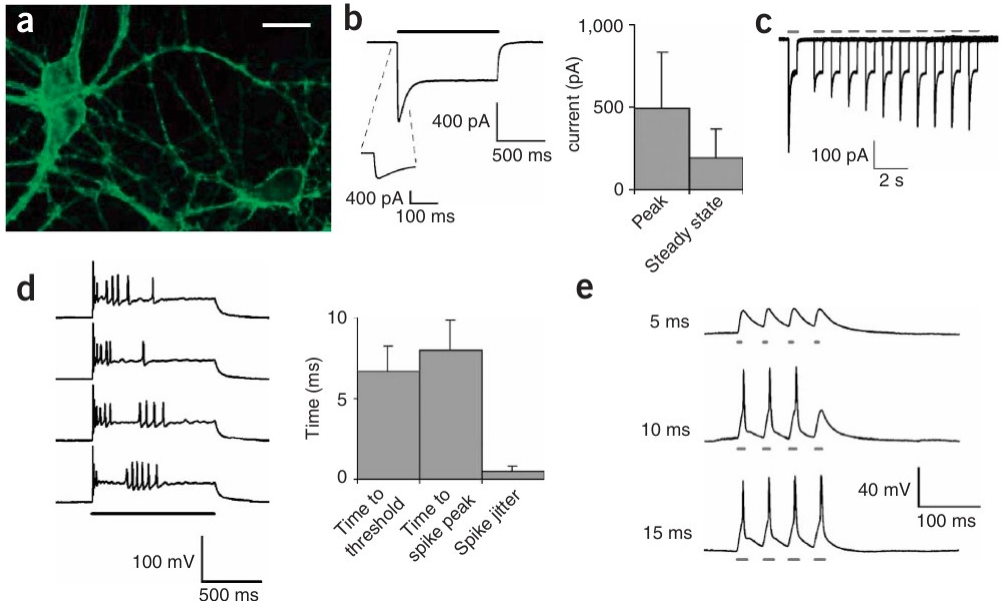
Nagel et al. (2003), Fig. 3(a–g). Excised-patch responses to light and intracellular pH.

Membrane expression enables optical drive of neurons

Boyden and colleagues introduced a ChR2 fusion with yellow fluorescent protein into cultured rat neurons using a **lentiviral vector**, a modified virus that delivers an expression construct. The fluorescent tag identified the expressing cells and their membrane-associated processes.

Blue light spanning 450–490 nm generated an average peak inward current near 496 pA, followed by a smaller sustained current near 193 pA. Neurons expressing the fluorescent tag alone did not produce light-evoked responses.

Inward current **depolarized** the neuron, making its internal voltage less negative. Reaching firing threshold triggered an **action potential**, a rapid regenerative change in membrane voltage. A **synapse** is a contact that transmits signals between cells. Optical stimulation provided control of neuronal firing and synaptic communication within the culture.



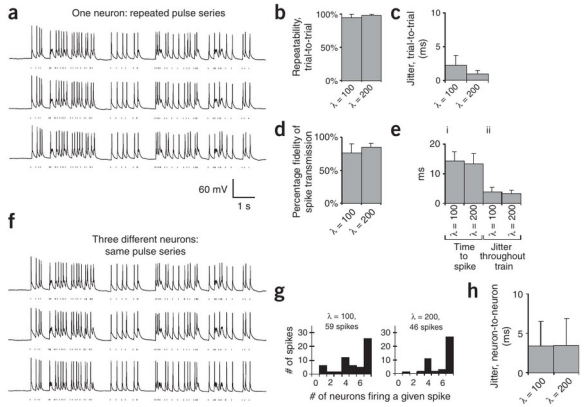
Boyden et al. (2005), Fig. 1(a–e). Membrane expression, inward current and light-evoked spikes.

Pulse timing controls spikes with finite reliability

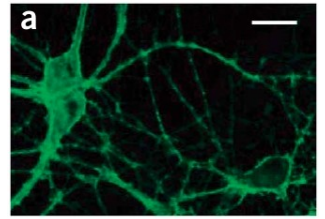
An **action potential** is a rapid regenerative change in membrane voltage that carries a neuronal signal. Boyden and colleagues recorded neurons while applying repeated 10–15-ms light pulses, comparing the timing of evoked spikes across repeated stimulation sequences.

The same pulse sequence produced highly repeatable spike patterns, but not every light pulse elicited a spike. In two irregular stimulation protocols, about 76% and 85% of pulses successfully drove spikes. Spike reliability declined at higher rates in tests spanning 5–30 Hz.

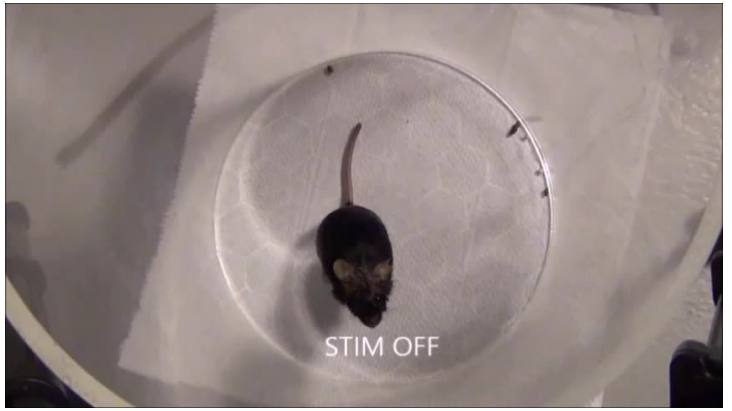
Continuous illumination produced a well-timed initial spike followed by activity with less precise timing. Brief repeated light pulses organized firing more closely around the stimulus times. In a later wireless experiment, Montgomery and colleagues delivered 20-Hz, 5-ms light pulses near premotor cortex. Illumination increased circling from approximately 0.40 to 2.5 turns per minute in mice expressing ChR2.



Boyden et al. (2005), Fig. 2(a–h). Repeated pulse trains and timing of evoked spikes.



Boyden et al. (2005), Fig. 1(a). ChR2-YFP membrane expression in a cultured hippocampal neuron.



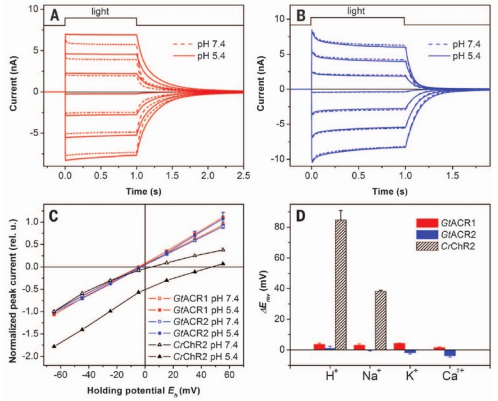
Montgomery et al. (2015), Supplementary Video 1. Wireless light delivery to premotor cortex and circling behavior.

Light can open channels that conduct chloride

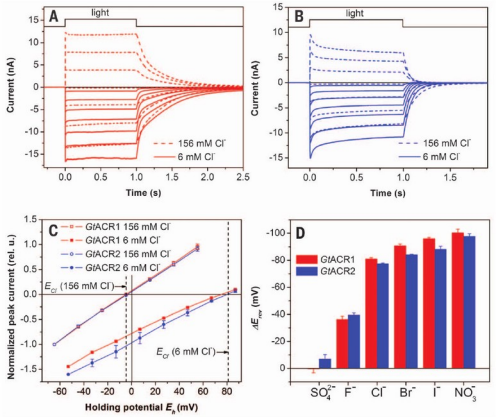
Govorunova and colleagues expressed light-sensitive proteins from the cryptophyte alga *Guillardia theta* in mammalian cells. Two proteins, GtACR1 and GtACR2, functioned as **anion channelrhodopsins**, light-gated channels carrying negatively charged ions rather than the positive ions carried by ChR2.

Changing external sodium, potassium or calcium did not shift photocurrent reversal as expected for cation conduction. Replacing chloride with a nonpermeant anion did shift reversal. Currents were measured while membrane voltage varied in 20-mV steps, with ion substitutions providing the selectivity controls.

The currents changed when chloride availability changed. Chloride movement therefore supplied the charge passing through the activated channels. The recorded current's direction depended on both the chloride gradient and membrane voltage, connecting an ion's distribution with the electrical response to light.



Govorunova et al. (2015), Fig. 2(A–D). Cation substitution and photocurrent reversal in GtACRs.



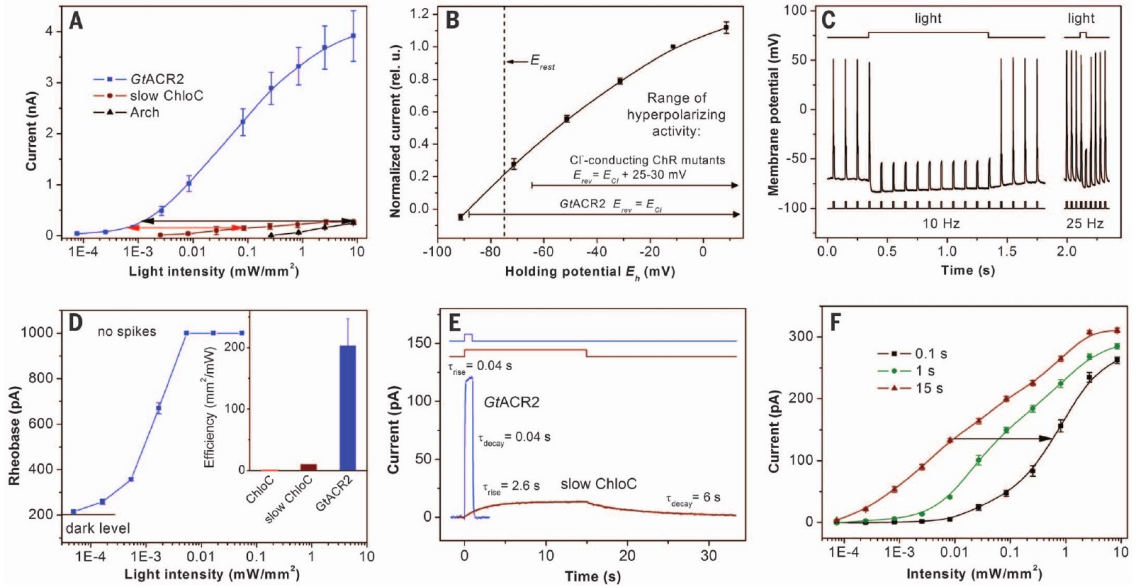
Govorunova et al. (2015), Fig. 3(A–D). Chloride substitution identifies anion selectivity.

Chloride movement changes neuronal excitability

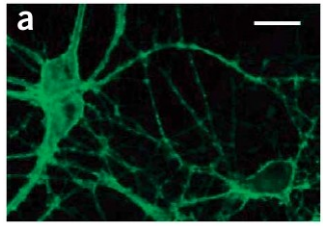
A **chloride equilibrium potential** is the voltage at which the electrical and concentration forces on chloride balance. Opening an anion channel moves the membrane response toward this voltage; the direction of the voltage change follows the ionic conditions in that cell.

In cultured rat neurons studied by Govorunova and colleagues, GtACR2 photocurrent reversed near −88 mV. Illumination suppressed current-evoked spikes, including trains up to at least 25 Hz. In the tested ramp protocol, full suppression occurred at a light intensity of 0.005 mW/mm².

During illumination, the anion channels conducted chloride across the neuronal membrane. Under the study's recording conditions, this current suppressed the generation of spikes. Turning the light on and off controlled the periods of reduced electrical activity, giving optical inhibition a measurable time course.



Govorunova et al. (2015), Fig. 4(A–F). GtACR2 current reversal, spike suppression and light sensitivity.



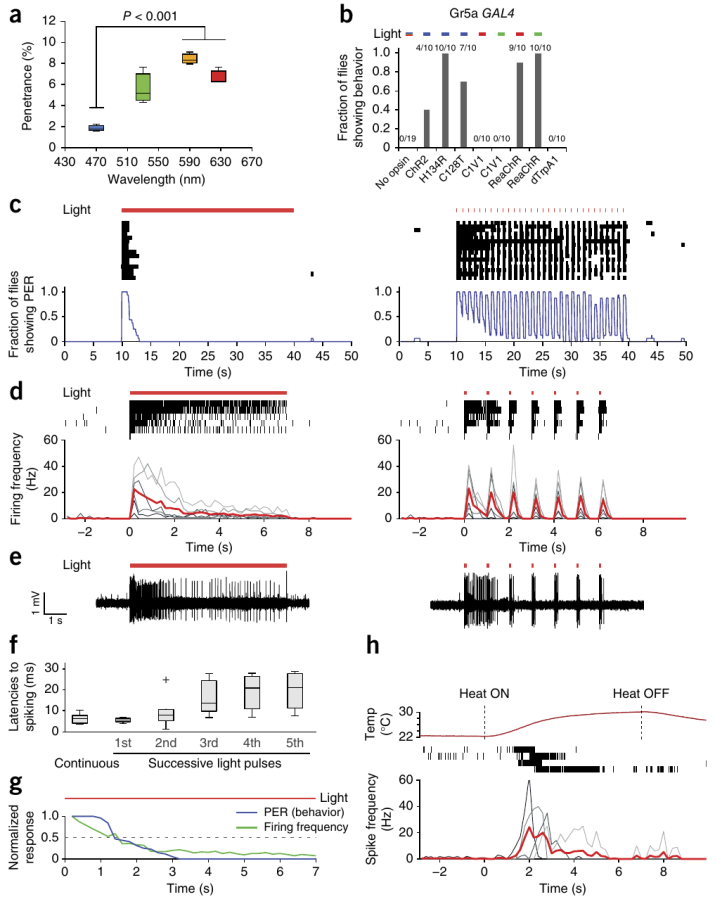
Boyden et al. (2005), Fig. 1(a). ChR2-YFP membrane expression in a cultured hippocampal neuron.

Red-shifted activation reaches neurons in intact flies

The adult fly's **cuticle**, its external body covering, limits light reaching internal neurons. Inagaki and colleagues measured light transmission and compared several **opsins**, retinal-binding light-sensitive proteins, in peripheral and central neuronal populations of intact *Drosophila melanogaster*.

The red-activatable channelrhodopsin **ReaChR** supported stimulation with 627-nm light. In gustatory receptor neurons, cells that detect chemical taste cues, 100-ms pulses at 1 Hz produced temporally controlled spiking and proboscis extension, the outward movement of the feeding appendage.

ReaChR also recruited behavioral responses through central neurons. Light first had to pass through the body covering and reach cells expressing the protein. Delivery of the opsin to the cell membrane then linked red illumination with electrical activity and the coordinated movement of the feeding or courtship system.



Inagaki et al. (2014), Fig. 1(a–h). Opsin comparisons, cuticle transmission and gustatory-neuron responses.



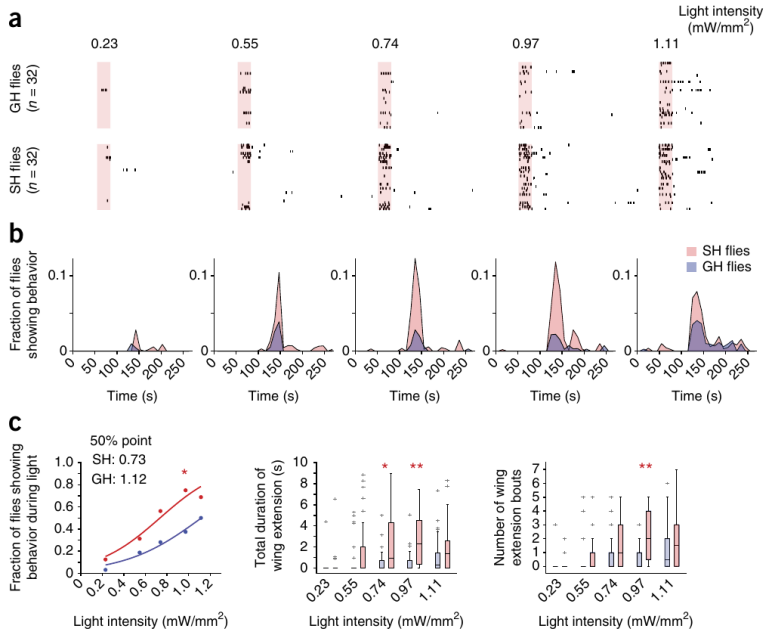
Grover et al. (2020), Fig. 1(c). Head imaging window and retroreflective beads on an adult fly.

Social history changes the effect of P1 activation

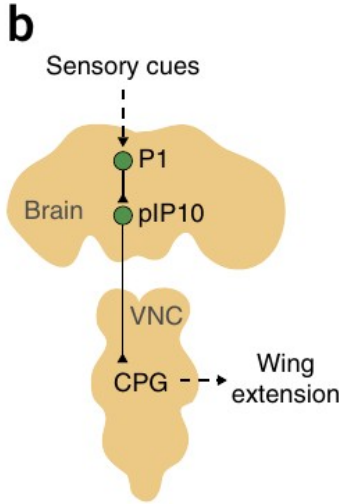
P1 neurons are a male-specific group of brain interneurons implicated in courtship control. An interneuron connects neural processing stages rather than directly serving as a muscle's motor output. Inagaki and colleagues compared ReaChR-driven wing extension after 7 days of single or group housing.

Single-housed males responded at lower red-light intensities than group-housed males. By contrast, activating the descending courtship neurons pIP10 did not reveal the same housing-dependent sensitivity shift. The comparison used matched stimulation methods while changing the animal's social experience.

The housing effect appeared when P1 neurons were activated, while activation of pIP10 output neurons produced a different sensitivity pattern. These functional comparisons placed the experience-related change within the pathway preceding the pIP10-dependent output. Social experience therefore changed how particular stages of the courtship system responded to an imposed neural input.



Inagaki et al. (2014), Fig. 4(a–c). Social housing changes P1-evoked wing-extension sensitivity.



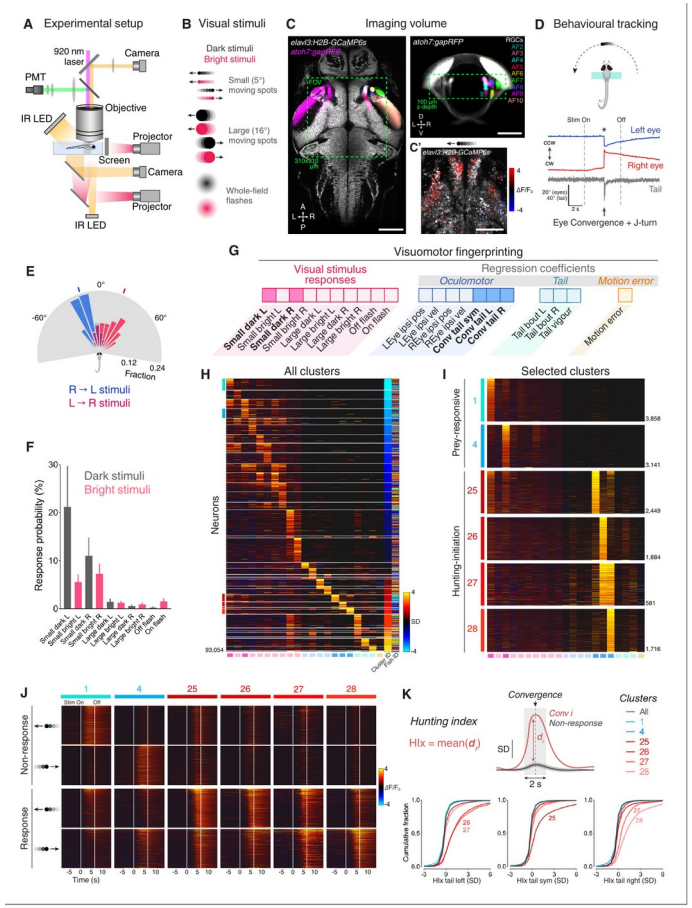
Inagaki et al. (2014), Fig. 3(b). Published P1 and pIP10 circuit arrangement; connectivity is proposed.

Hunting begins with coordinated eye and tail movements

Larval zebrafish hunt small moving prey using a coordinated sequence of eye and body movements. **Eye convergence** rotates the eyes toward a common frontal region. A **saccade** is a rapid eye movement; convergent saccades provide a measurable marker of hunting initiation.

Antinucci and colleagues studied larvae 6–7 days after fertilization. During a partially restrained visual assay, small dark moving spots preferentially recruited convergence together with lateral tail movement. Approximately 92% of convergent saccades coincided with tail motion, with an average eye-to-tail delay near 12 ms.

Calcium imaging follows activity-related changes in a fluorescent indicator inside cells. Two-photon microscopy uses focused light to excite that indicator within the tissue. Recording neural fluorescence together with eye and tail movement connected a moving visual stimulus with both the brain's response and the coordinated onset of hunting.



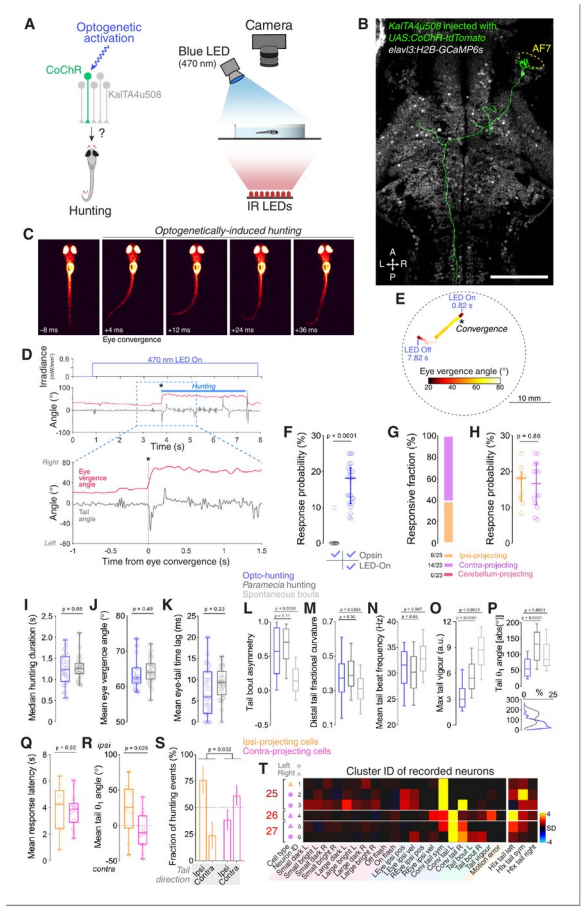
Antinucci et al. (2019), Fig. 1(A–K). Hunting-related calcium activity, eye convergence and brain anatomy.

Activating one pretectal cell can recruit hunting

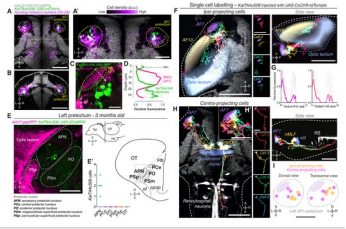
Antinucci and colleagues expressed **CoChR**, a light-gated cation channel, in individual genetically accessible neurons in the **pretectum**, a visual-processing region near the midbrain. Cell morphology identified the targeted neuron, and high-speed video quantified free-swimming responses during repeated 7-s blue-light stimulation periods.

Illumination at 470 nm and 0.44 mW/mm² caused some single-cell preparations to produce eye convergence followed by lateralized swim bouts and sustained hunting-like routines without prey. Responsive preparations had a median response probability near 18% during illumination, compared with zero in the corresponding light-off trials.

A single-cell intervention could recruit eye convergence, directed swim movements, and a sustained hunting-like sequence. **Sufficiency** means that activating the tested element can produce the measured response under the experimental conditions. The preparation linked activity in an identified neuron with the initiation of several coordinated movement components.



Antinucci et al. (2019), Fig. 5(A–T). Single-neuron optical activation and hunting-like kinematics.



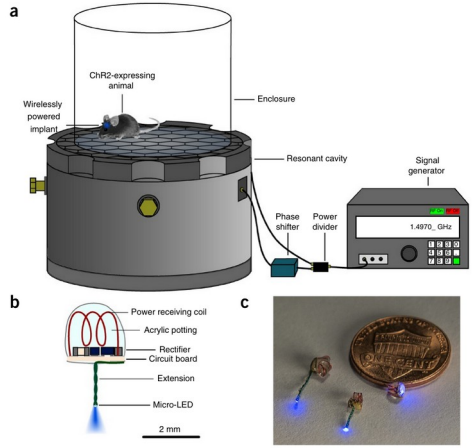
Antinucci et al. (2019), Fig. 3(A–I). Single pretectal-cell morphology and projection classes.

An internal implant can receive wireless optical power

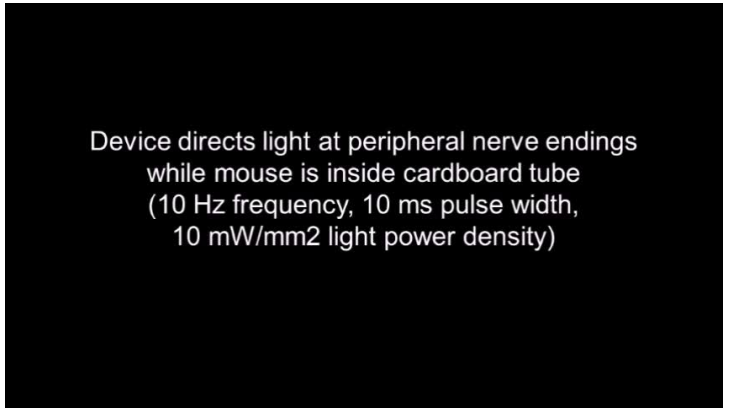
Montgomery and colleagues built fully internal light-delivery implants for mice. A **micro-LED** is a miniature light-emitting diode; it converts electrical power into light near the targeted tissue. A receiving coil and local electronics obtained power from an external radio-frequency apparatus.

The smallest implant weighed about 20 mg and occupied about 10 mm³. An aluminum resonant cavity operating near 1.5 GHz concentrated electromagnetic energy in the mouse, powering a blue LED without an external fiber or battery attached to the animal.

The implanted receiver converted incoming radio-frequency energy into electrical power for a light-emitting diode. The diode delivered light near the selected neural target. Separating the external power source from the internal optical output enabled stimulation while the mouse moved through the arena.



Montgomery et al. (2015), Fig. 1(a–c). Wireless-power preparation and miniaturized implanted light sources.



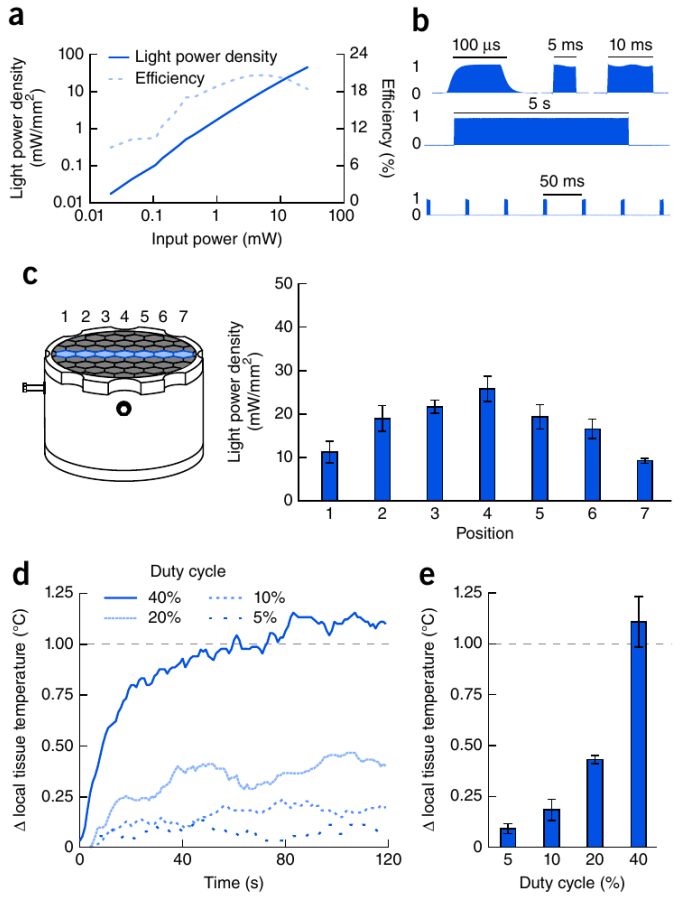
Montgomery et al. (2015), Supplementary Video 2. Wireless light delivery during running, tube exploration, and social interaction.

Wireless optical stimulation delivers pulsed energy

Electrical energy powers the light emitter and can warm the surrounding tissue. Montgomery and colleagues calibrated the LED's emitted power, pulse fidelity and position-dependent power delivery, using both controlled electrical tests and measurements in a mouse-sized saline preparation.

The device produced pulses as short as 100 µs. At a fixed cavity input, estimated light power density across the tested surface varied from approximately 9.2 to 25.8 mW/mm². Brain-adjacent measurements with 20 mW/mm² illumination remained below a 1 °C temperature rise at duty cycles up to 20%.

Duty cycle is the fraction of time that stimulation is on. Increasing duty increased the measured heating, with a rise exceeding 1 °C near the tested 40% condition. Pulse timing therefore changed both the pattern of delivered illumination and the average energy deposited around the device.



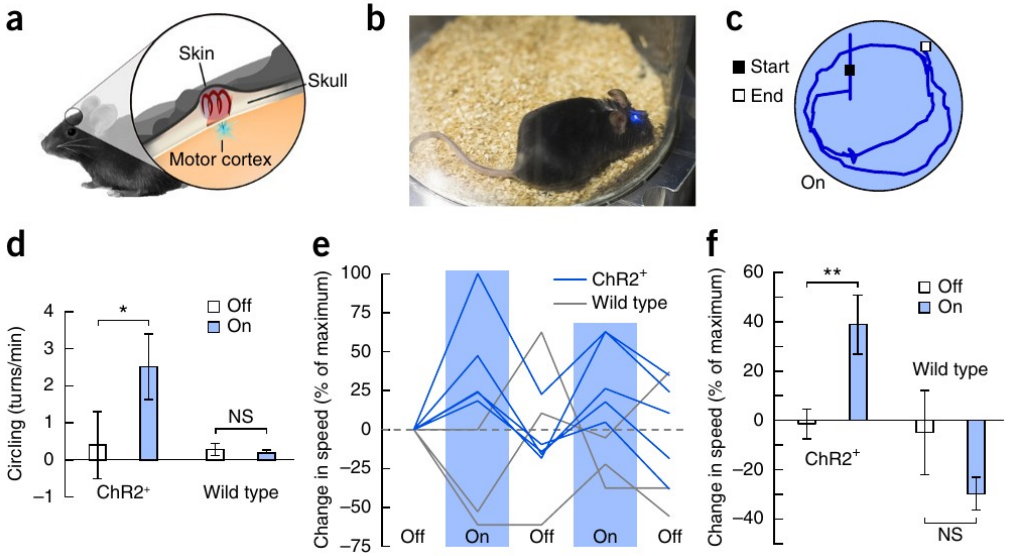
Montgomery et al. (2015), Fig. 2(a–e). LED power, pulse fidelity, position dependence and tissue heating.

Wireless cortical stimulation changes turning behavior

Montgomery and colleagues implanted a wireless LED near the right **premotor cortex**, a cortical region participating in movement control. ChR2-expressing mice and implanted wild-type controls were tested in the same powered arena while video quantified movement.

During 20-s stimulation epochs, 5-ms pulses at 20 Hz increased circling from approximately 0.40 to 2.5 turns/min in the ChR2 group. Mean normalized locomotor speed increased by about 40%. Wild-type controls did not display the corresponding light-dependent movement changes.

Wireless illumination changed the mouse's trajectory during stimulation of the premotor region. The animal could move without a light-delivery cable attached to the external apparatus. A radio-powered optical input therefore produced a temporally controlled change in movement through neurons expressing the light-sensitive channel.



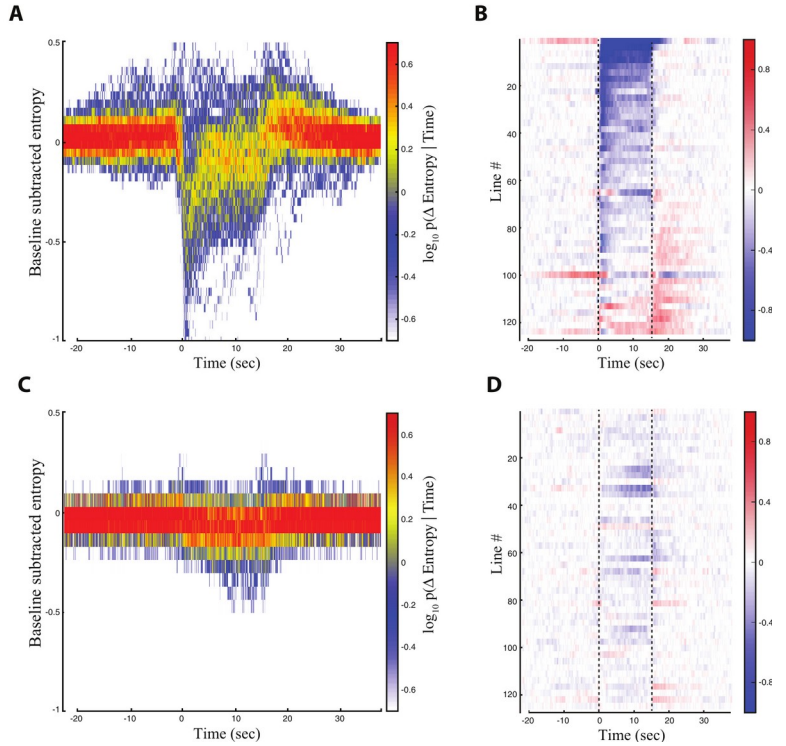
Montgomery et al. (2015), Fig. 3(a–f). Implanted premotor-cortex stimulation and circling behavior.

Descending activation often favors one motor module

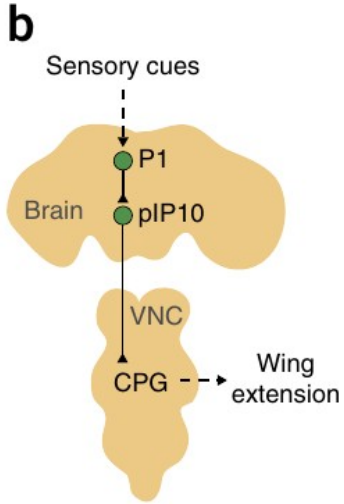
Descending neurons carry signals from the fly's brain toward motor regions in its ventral nerve cord. Cande and colleagues activated selected groups using a red-light-sensitive protein. Video measurements recorded the resulting movements in the experimental arena.

The movements were classified into coordinated behavioral patterns. A **motor module** is a coordinated component of behavior. During 15-second illumination periods, many activated groups shifted the animal's activity toward a particular behavioral region, changing which pattern occupied most of its time.

Different targeted groups produced different distributions of movement. Many effective groups predominantly favored one region of the measured behavioral repertoire. The experiment connected genetically identified neural populations with shifts in the coordinated components of ongoing behavior.



Cande et al. (2018), Fig. 3(A–D). Behavioral entropy across descending-neuron lines.



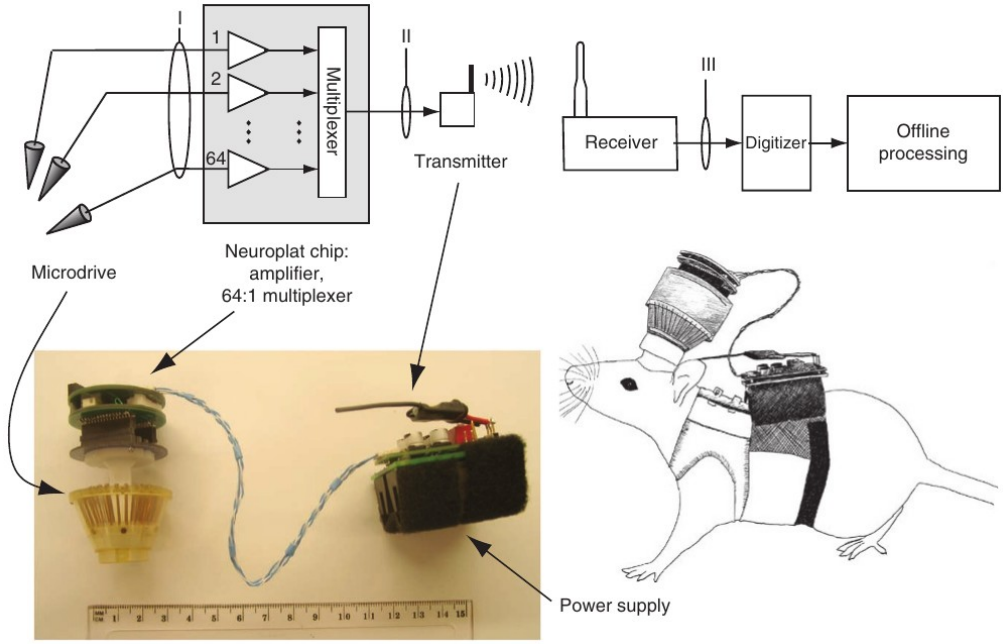
Inagaki et al. (2014), Fig. 3(b). Published P1 and pIP10 circuit arrangement; connectivity is proposed.

Wireless telemetry removes the external recording tether

Neural telemetry transmits recorded electrical signals remotely. Szuts and colleagues built a rat-borne system combining implanted electrodes, a head-mounted amplifier and a radio transmitter carried in a back harness. A short connection remained between the animal's own boards, but no cable linked the rat to the recording computer.

The electronics could process up to 64 channels sampled at 20 kHz each and send the combined signal to a receiver. Continuous operation lasted approximately 6 h with the larger battery configuration. The arrangement enabled recordings during outdoor movement and travel through tunnels.

The head-mounted amplifier increased the small electrical signals measured by implanted electrodes. Electronics converted the measurements into a stream transmitted by the animal-borne radio. The external receiver reconstructed and stored that stream, connecting activity recorded during movement with measurements analyzed on a computer.



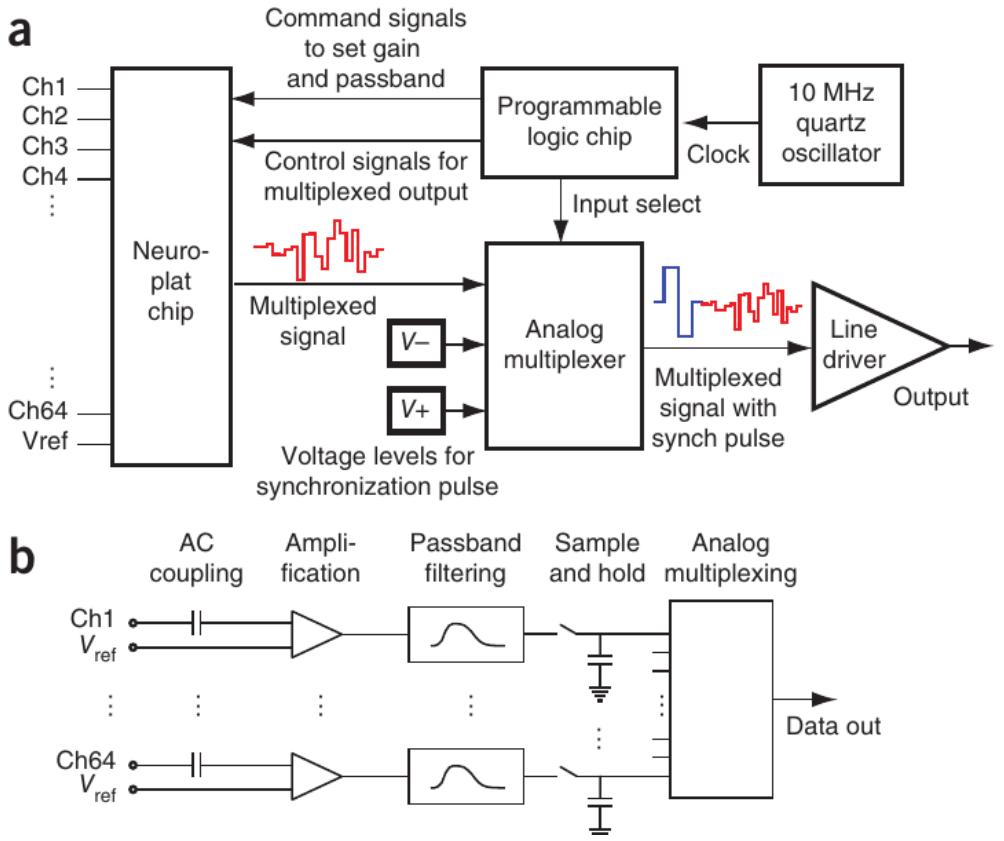
Szuts et al. (2011), Fig. 1. Published wireless headstage, harness and recording components.

Recording systems sample changing neural signals

Sampling rate specifies how often a signal is measured, whereas **bandwidth** specifies the range of frequencies retained by the system. Szuts and colleagues filtered electrode voltages from 80 to 2,300 Hz and sampled each channel at 20 kHz in their recording configuration.

A simultaneous sample-and-hold stage captured the channel voltages before **time-division multiplexing**, placing successive channels onto a shared output. Synchronization markers identified each channel scan so that the original signals could be reconstructed after radio transmission.

Extracellular spikes are brief voltage changes measured near active neurons. The selected filter range retained frequencies useful for identifying those changes. Repeated sampling and synchronization markers preserved the timing and ordering needed to reconstruct each electrode's signal after transmission.



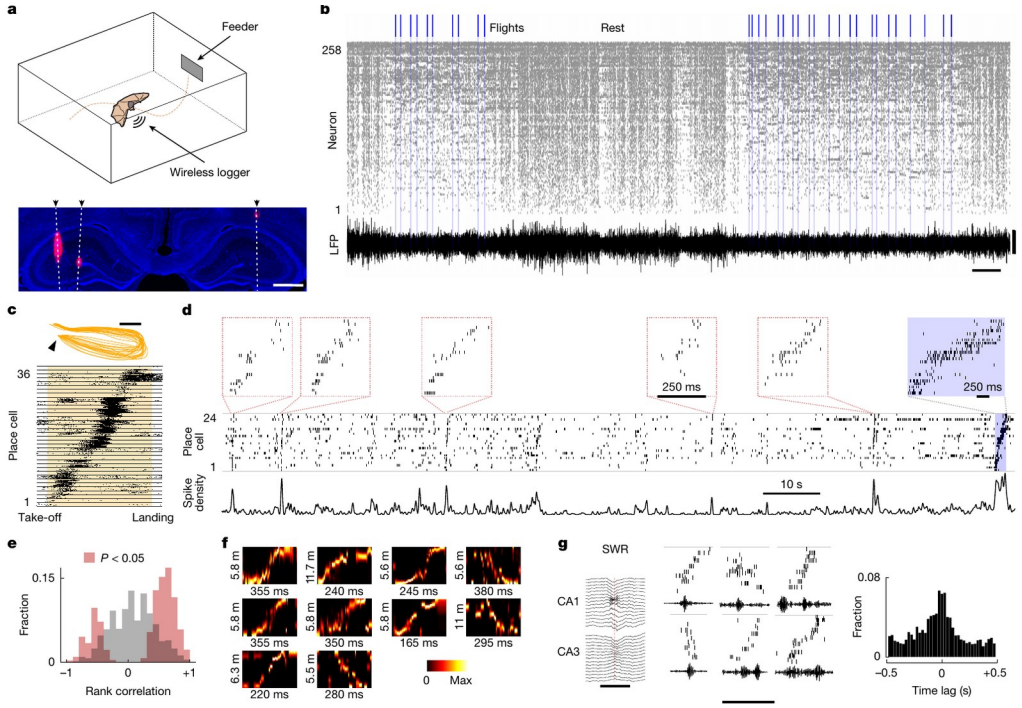
Szuts et al. (2011), Fig. 2(a–b). Sampling, multiplexing and signal conditioning in the original device.

Neural activity can be recorded during free flight

A **neural ensemble** is a population of neurons whose combined activity can represent information. Forli and colleagues used wireless **Neuropixels recordings**, using a probe with many densely packed recording contacts, from the dorsal **hippocampus**, a brain region involved in spatial learning and memory, of freely flying Egyptian fruit bats engaged in self-paced, rewarded aerial foraging.

The system recorded **single-unit activity**, signals assigned to individual neurons, together with **local field potentials**, extracellular signals reflecting nearby population currents. Flights alternated spontaneously with rest. About 71% of flight-active neurons met the study's criterion for spatial selectivity across the sampled trajectories.

Recording a population allowed the researchers to compare the order in which different cells became active during a flight. Spatially selective neurons discharged at different locations along the trajectories. Their combined activity provided a changing representation that could be compared with the bat's measured position and movement.



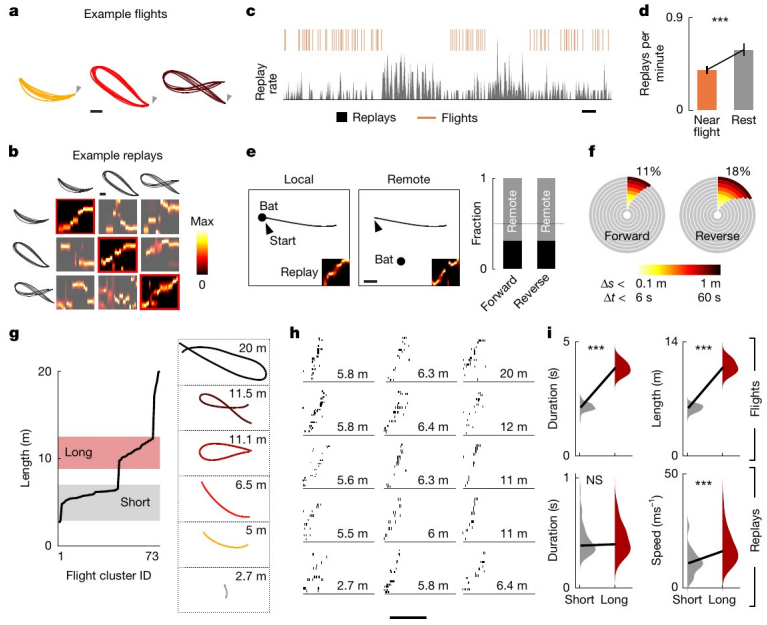
Forli et al. (2025), Fig. 1(a–j). Wireless Neuropixels recordings and flight-sequence replay.

Rest events replay compressed flight sequences

Replay is the reactivation of an ordered neuronal sequence related to an earlier behavioral trajectory. Forli and colleagues compared hippocampal place-cell activity during flight with brief population events occurring while the bats rested. **Place cells** are neurons with position-dependent discharge.

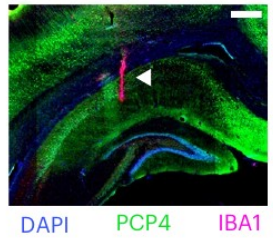
Detected replay events had a median duration near 358 ms and could preserve either the forward or reverse ordering of flight-related activity. They coincided with **sharp-wave ripples**, brief coordinated field-potential events, and frequently represented paths remote from the bat's current resting location.

Decoded replay trajectories frequently represented locations away from the bat's resting position. The animal's body could remain stationary while its hippocampal activity passed through an ordered representation of another trajectory. Brief rest events therefore contained structured spatial information associated with flight-related activity sequences.



Forli et al. (2025), Fig. 2(a–i). Replay location, timing and scaling across flight trajectories.

d



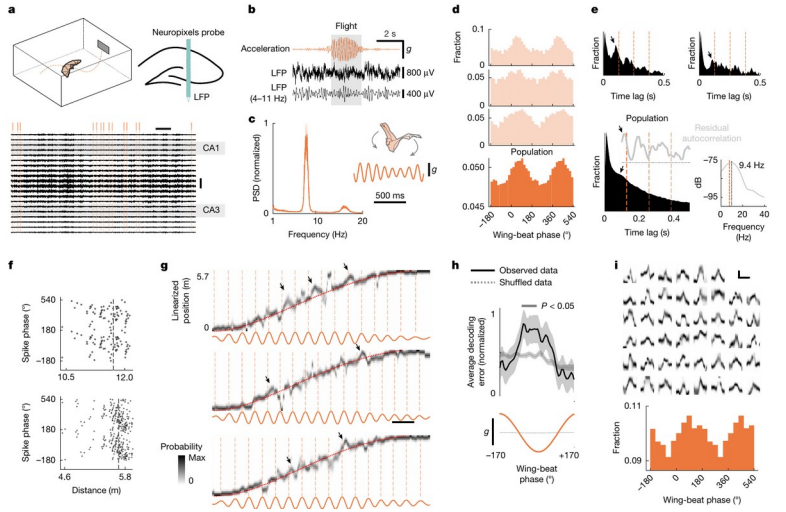
Snyder et al. (2024), Fig. 1(d). Recorded dorsal hippocampus with electrode tracks and a 500-µm scale bar.

Flight-related representations track a motor rhythm

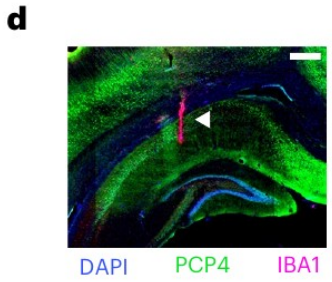
Forli and colleagues recorded hippocampal activity alongside the movements of freely flying bats. **Place cells** have firing patterns that vary with the animal's position. Accelerometers recorded the repeated body movements associated with wing beats during flight.

The wing-beat rhythm was prominent near 8 Hz. Many place cells fired preferentially during particular parts of that cycle. **Phase** describes a point within a repeating cycle, allowing neuronal firing times to be compared with the timing of movement.

Decoding uses patterns of neural activity to estimate the information represented by the population. Decoded position swept ahead of the bat's current location in repeated cycles. These **representational sweeps** changed in relation to the ongoing rhythm of the animal's movement.



Forli et al. (2025), Fig. 3(a–k). Representational sweeps and wing-beat phase during flight.



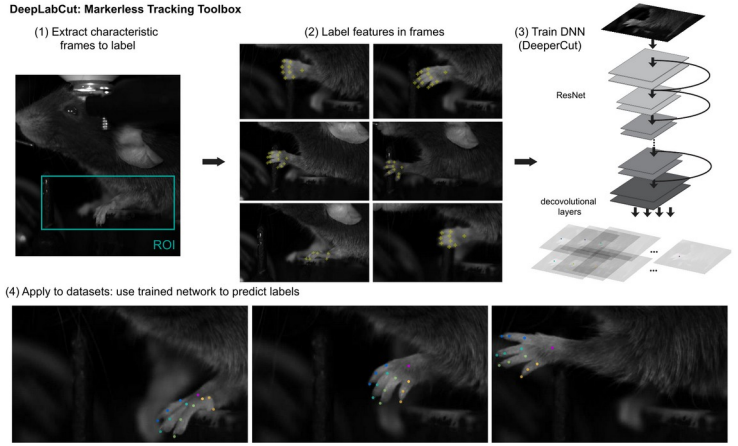
Snyder et al. (2024), Fig. 1(d). Recorded dorsal hippocampus with electrode tracks and a 500-µm scale bar.

Computer vision measures the position of body parts

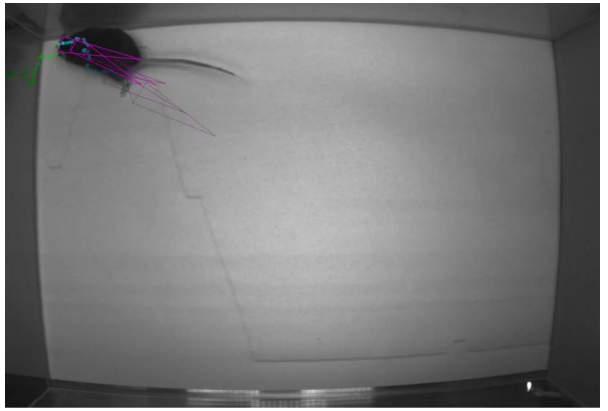
Computer vision uses computational methods to extract information from images. **Pose estimation** identifies the locations of body parts such as the snout, ears, wrist, or individual digits. A sequence of estimated positions describes how the animal's configuration changes over time.

Mathis and colleagues developed DeepLabCut to locate selected body parts in ordinary video. The user first identifies features of interest in example frames. The trained system then predicts the locations of those features in additional frames.

The method was applied to mice following an odor trail, flies moving within a chamber, and mice performing a reaching task. Different features were selected for each preparation. The same analysis framework turned recorded images into measurements tailored to each behavior.



Mathis et al. (2018), Fig. 1 (author preprint). Labeling, training, and markerless pose estimation.



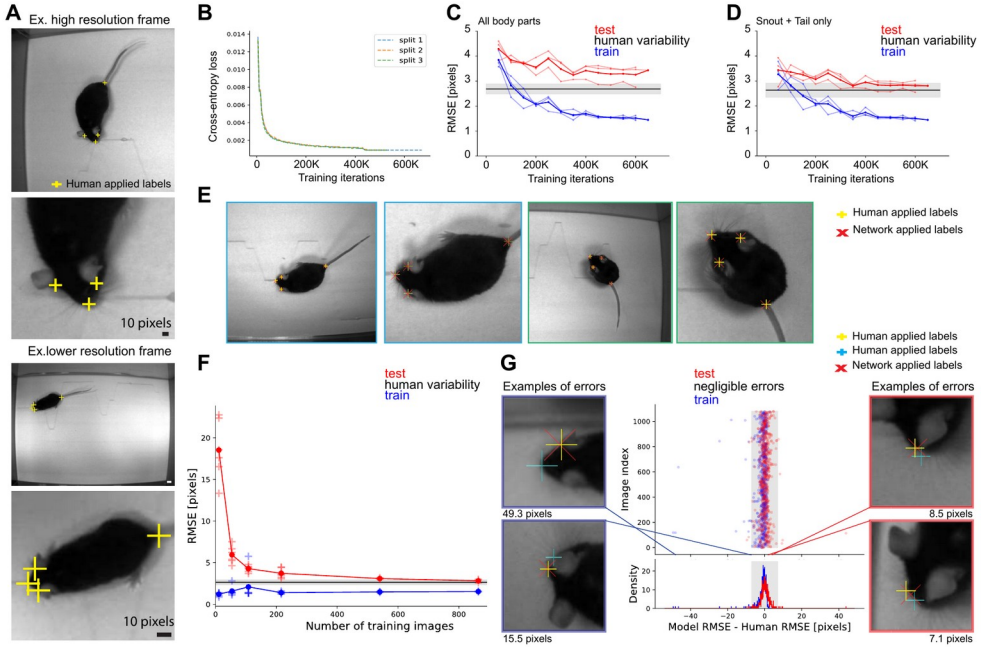
Mathis et al. (2018), Supplementary Video 1. Markerless tracking of a mouse following an odor trail.

Labeled images define anatomical landmarks

A **label** specifies the position assigned to a selected body part in an example image. Researchers selected frames containing characteristic postures and marked the locations of the features they wanted to follow. These paired images and labels supplied training information.

The image-processing network learned relationships between visual features and body-part locations. The authors trained detectors for features such as the snout and ears. Predictions could then be compared with locations assigned by a human observer.

Training used a small collection of labeled images rather than requiring every video frame to be labeled manually. The study reported accurate tracking using about 200 labeled frames in its examples. Once trained, the detector applied the learned relationship to a much longer video sequence.



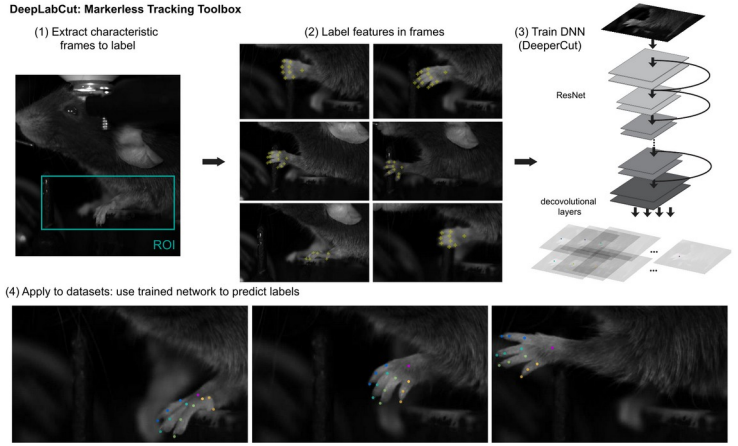
Mathis et al. (2018), Fig. 2A–G (author preprint). Tracking precision and training examples.

Transfer learning adapts visual features to a new task

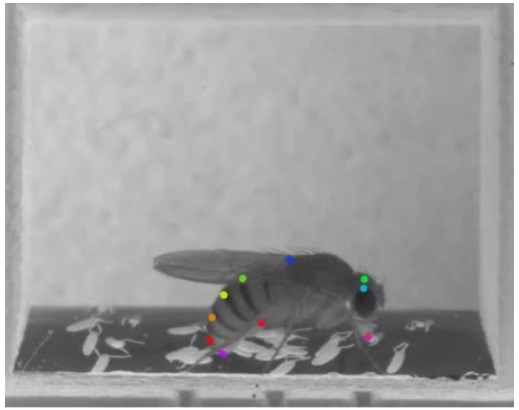
Transfer learning starts with information learned for one computational task and adapts it to another. DeepLabCut used image-processing features from a network previously trained on a large collection of images. The researcher then trained the system to locate body parts relevant to the experiment.

An **artificial neural network** is a computational system that transforms inputs through connected processing operations. In this application, the input is an image and the output identifies likely body-part positions. Training changes the computational parameters used to generate those outputs.

The authors adapted the approach to both mouse and fly behavior. Reusing previously learned visual features reduced the amount of task-specific labeling required. The resulting measurements followed anatomical landmarks chosen by the researcher.



Mathis et al. (2018), Fig. 1 (author preprint). Labeling, training, and markerless pose estimation.



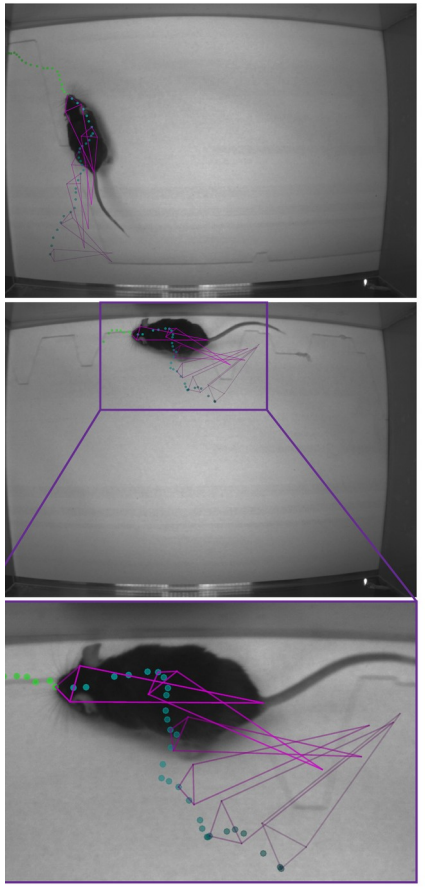
Mathis et al. (2018), Supplementary Video 2. Markerless tracking of fly body parts.

Tracking resolves head movements during odor following

During odor-trail following, the mouse moved its snout from side to side across the trail. DeepLabCut tracked the snout together with the ears and tail base. These positions described the movement of the head and the orientation of the body.

A **trajectory** is the sequence of positions occupied over time. Combining tracked positions across frames produced snout trajectories during the behavior. The recordings connected the animal's movement with the location of the odor trail.

The study also tracked flies, including body landmarks involved in positioning the head and abdomen. A **proboscis** is the fly's extendable feeding structure. Tracking its extension provided another behavioral measurement using the same computational approach.



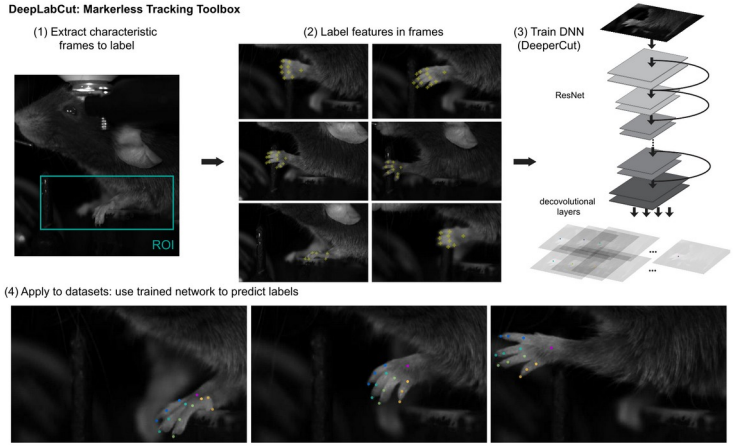
Mathis et al. (2018), Fig. 3 (author preprint). Snout and body trajectories during odor tracking.

Digit tracking resolves a skilled reaching movement

A reaching movement combines the motion of the forelimb, wrist, and digits. Mathis and colleagues used a mouse reaching preparation to track selected landmarks across the movement. Their examples included individual digit joints and the wrist.

The computer-generated positions allowed the movement to be described at a finer anatomical scale than a single whole-body location. Changes in several landmarks could be measured in the same recorded sequence. The detected positions described the configuration of the hand as the task unfolded.

Body-part selection determined the behavioral information extracted from the video. Snout positions supported odor-navigation measurements, while digit positions supported a reaching measurement. Selecting landmarks tied the computational output to the movement that the experiment investigated.



Mathis et al. (2018), Fig. 1 (author preprint). Labeling, training, and markerless pose estimation.



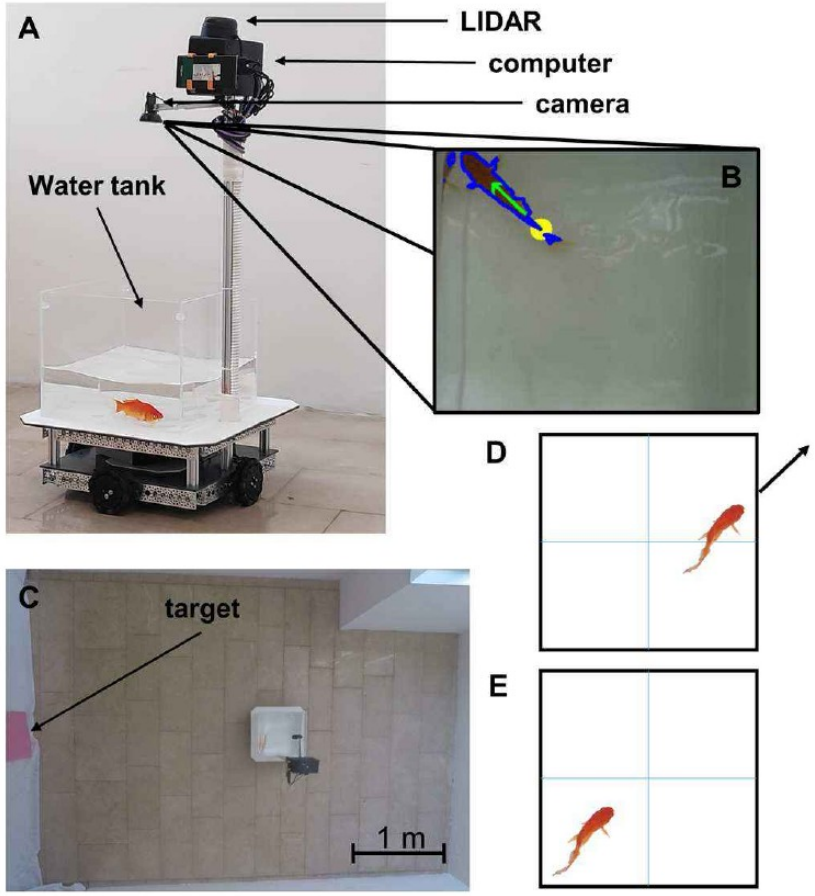
Mathis et al. (2018), Supplementary Video 3. Mouse digit and wrist tracking during a reaching movement.

Goldfish drive toward a terrestrial visual target

Goldfish can control a wheeled platform while remaining in water. Givon and colleagues placed a water tank on a **fish-operated vehicle**, or FOV, whose movement depended on the fish's position and orientation. The animal's swimming actions displaced its tank through a room.

A pink board marked the rewarded destination in a 3 × 4 m arena. Reaching it produced a food pellet. With training, fish reached the target more often and followed more direct routes, combining control of an unfamiliar device with visual guidance toward a goal.

Domain transfer means performing a familiar class of task in a different environmental setting. Here, aquatic movements controlled terrestrial translation. Successful navigation required learning how movements inside the tank changed the animal's position relative to objects outside it.



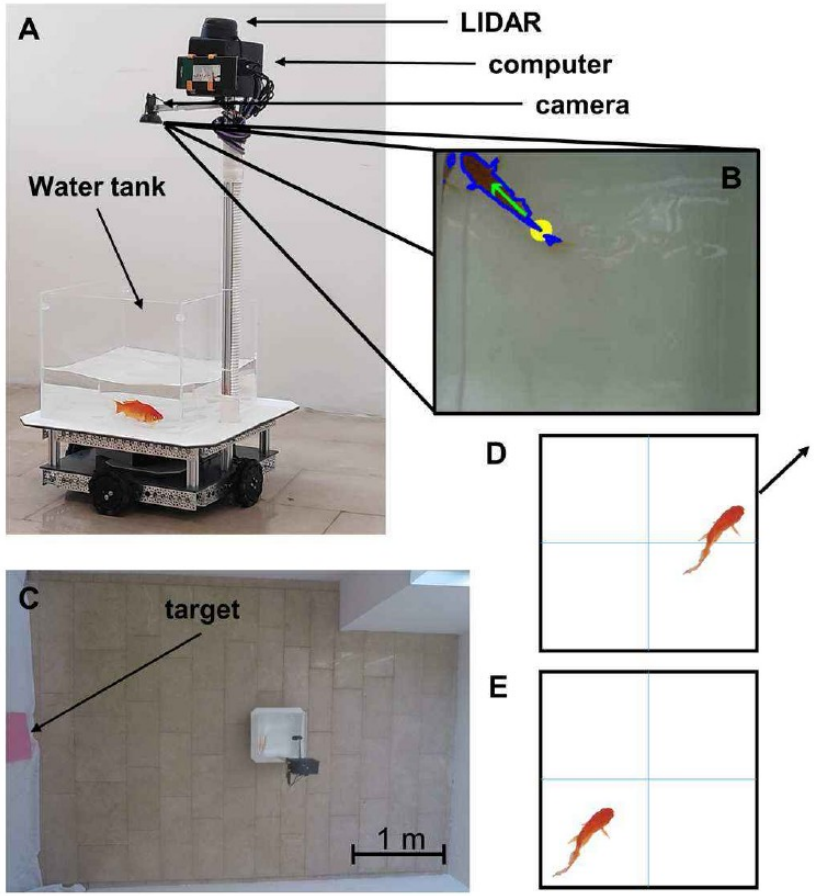
Givon et al. (2022), Fig. 1(A–E). The fish-operated vehicle and its control mapping.

Tracking links fish position to vehicle movement

The vehicle’s camera viewed the goldfish from above and continuously recorded its location inside the tank. A computer-vision algorithm separated the fish from the background, identified its body contour, and estimated the direction it faced. These measurements supplied the commands sent to the wheel motors.

The control arrangement made the fish’s body position behaviorally relevant. Swimming toward a tank wall while facing outward could move the entire platform in that direction. The same movement therefore changed both the fish’s local position and its relationship to a distant target.

The investigators recorded the fish’s tank position separately from the vehicle’s room position. This separation distinguished the animal’s controlling actions from the path those actions produced. Learning could consequently be assessed through target arrivals, movement duration, and the vehicle’s route through the arena.



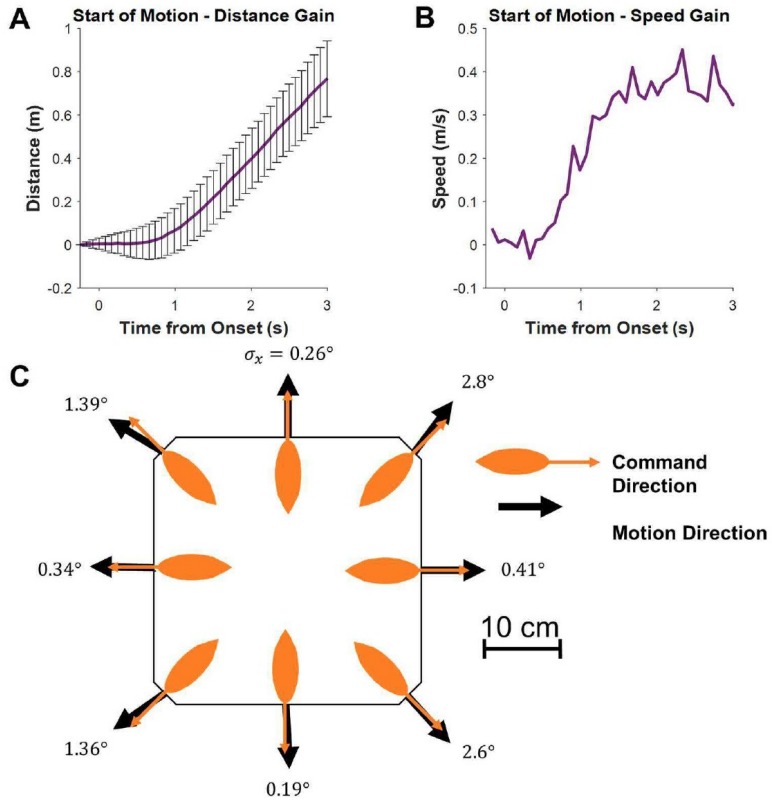
Givon et al. (2022), Fig. 1(A–E). The fish-operated vehicle and its control mapping.

Orientation and tank position control the wheels

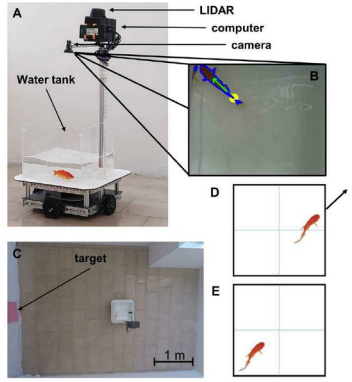
Vehicle movement depended on a conjunction of location and orientation. The goldfish had to approach a tank boundary and face outward before its action produced translation. Facing inward or remaining away from the boundary could leave the platform stationary, so swimming alone did not guarantee progress toward the target.

Motor measurements established how commands changed vehicle motion. The response developed over approximately one second, rather than instantly. The fish therefore experienced a delay between changing its controlling position and observing the resulting displacement through the room.

A laser-ranging system also prevented the platform from moving closer than 20 cm to an obstacle. Collision prevention was part of the device's control system. The learned task combined fish-directed movement with mechanical delay and automatic restrictions near walls, which together shaped the available routes.



Givon et al. (2022), Fig. 2(A–C). Motor delay and directional response of the vehicle.



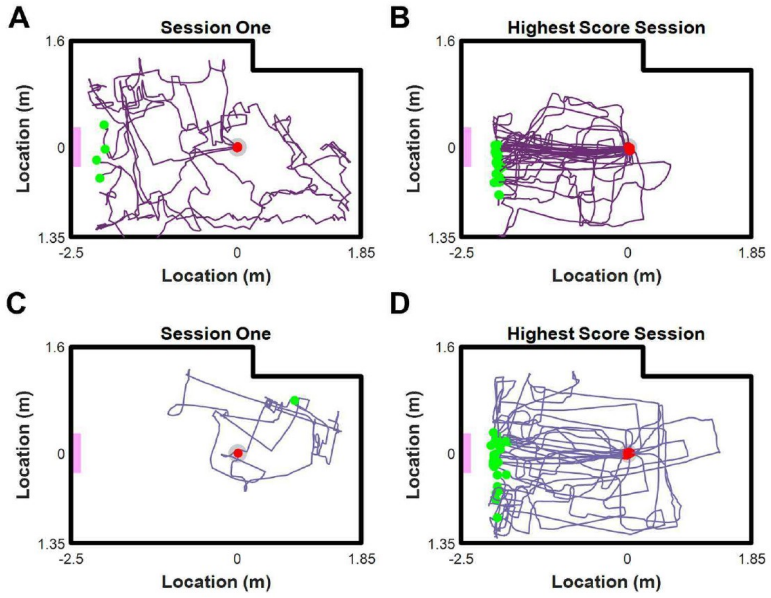
Givon et al. (2022), Fig. 1(A–E). The fish-operated vehicle and its control mapping.

Rewarded training produces more direct routes

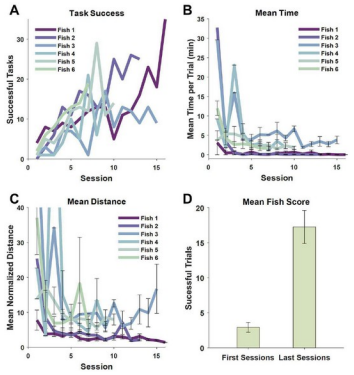
Operant learning changes behavior through its consequences. In the vehicle task, a food pellet followed arrival at the pink target board. The experimenters returned the covered tank to the starting position, then allowed another attempt, linking successful vehicle control with a repeated reward.

Early routes wandered through the arena and often failed to reach the destination. Later routes were more direct. Training increased successful target arrivals while reducing travel time and path length, so the improvement concerned both obtaining reward and organizing movement toward its location.

Sessions lasted up to 30 minutes and ended after the trial limit was reached. Comparing routes across sessions tracked acquisition of the action-to-motion relationship. More efficient navigation emerged after repeated experience with the same device and visual goal, rather than from an immediately available terrestrial motor repertoire.



Givon et al. (2022), Fig. 3(A–D). Routes become more direct with training.



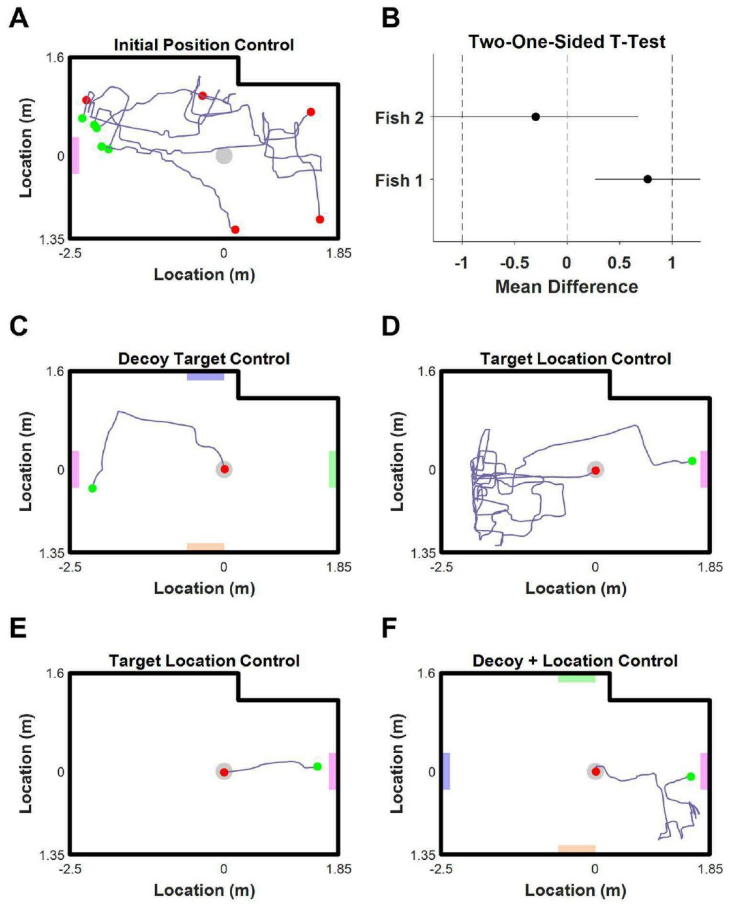
Givon et al. (2022), Fig. 4(A–D). Training improves target arrivals, travel time, and path length.

Changed starting positions test route flexibility

A fixed starting position permits a repeated sequence of movements to reach a goal. Givon and colleagues moved the vehicle to different starting locations after training, changing the route required to reach the same target. Successful transfer therefore required adapting the trajectory to the current situation.

Fish reached the destination from the new positions. The researchers compared route efficiency relative to the shortest path from each starting point, because raw travel distance necessarily changes when a start moves. Performance was compatible with continued goal-directed navigation under the altered starting conditions.

A **reference frame** specifies the coordinates used to describe a location or movement. A learned route tied only to the original start would become inappropriate after displacement. The new-start trials instead required the fish to relate the vehicle's present position to the target visible in the surrounding arena.



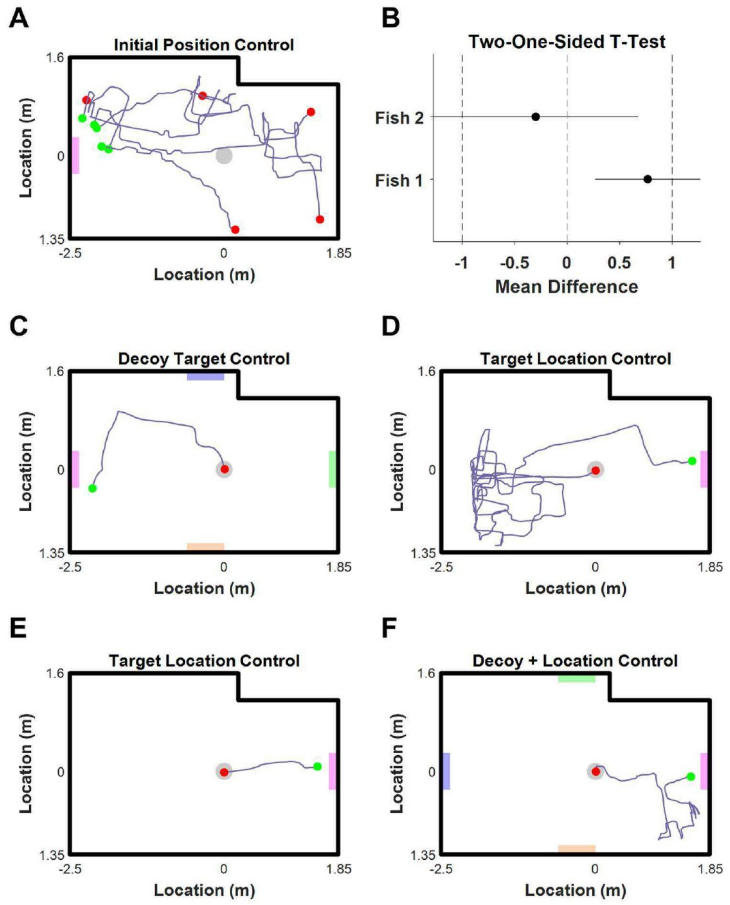
Givon et al. (2022), Fig. 5(A–F). Changing starts, decoys, and relocated targets tests navigation.

Decoys and target relocation test visual guidance

The target board had both a distinctive appearance and a stable location during training. Same-shaped boards of other colors acted as **decoys**, competing objects that resembled the target without carrying its reward. Fish continued toward the pink destination rather than treating every rectangular board as equivalent.

Moving the pink board to the opposite wall separated its appearance from its original place. In the reported first encounter, the fish repeatedly approached the old location before reaching the relocated board. By the third trial, the route went directly toward the new target position.

A combined test changed the target location while retaining decoys. Successful approach under this arrangement linked target identity with an updated destination. The sequence of old-place visits and subsequent redirection indicates that learned place information and current visual information both contributed to behavior.



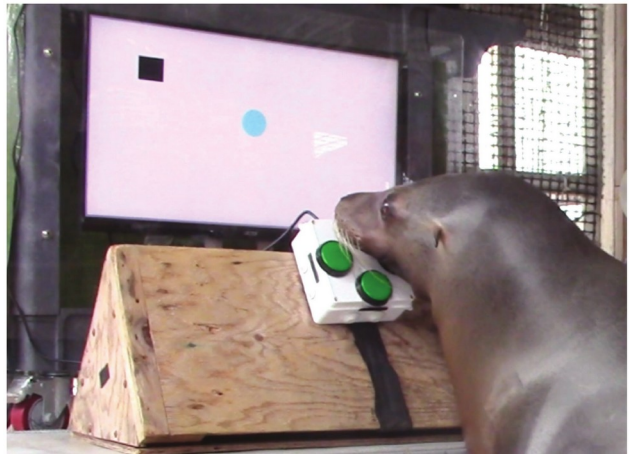
Givon et al. (2022), Fig. 5(A–F). Changing starts, decoys, and relocated targets tests navigation.

Sea lions learn to control a computer cursor

Cognitive enrichment provides opportunities to use perception, learning and decision making within an animal's managed environment. Winship and colleagues introduced Enclosure Video Enrichment, abbreviated **EVE**, to California sea lions housed in outdoor sea pens. The animals controlled a visible cursor rather than watching a prerecorded sequence.

Pressing directional buttons moved a blue circle toward a black target. Contact with the target produced an acoustic signal and reinforcement. This arrangement linked the animal's action to an observable consequence on the display and subsequently to a reward.

The sea lions progressed through successive training phases and acquired enough cursor control to enter additional games. Their movements therefore affected the ongoing task. Watching a screen, contacting a controller and directing a cursor are separate behavioral achievements.



Winship et al. (2023), Fig. 7. Sea lion operating the cursor task



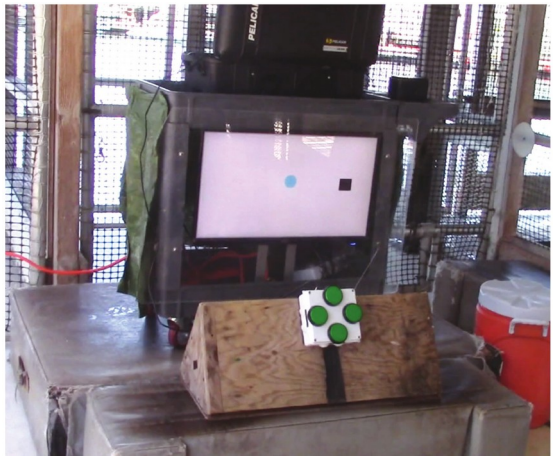
Winship et al. (2023), EVE excerpt 0:00–0:13. Brief button contact.

A display and a touchscreen are different interfaces

A **touchscreen** registers contact at a location on its display. EVE instead separated the display from the input device: a monitor presented visual targets, while four buttons controlled cursor movement. The sea lion selected movements rather than directly touching the target's displayed position.

This separation matters for task demands. A direct-contact response identifies a location, whereas cursor control requires a relationship between button direction, duration of contact and movement elsewhere in the visual field. The researchers trained that relationship explicitly rather than assuming it was already understood.

The sea lion used muzzle contacts to operate the directional buttons while monitoring the separate display. Each input moved the same visible cursor, and the animal coordinated button selection with the cursor's changing position. Reaching a target therefore combined visual monitoring with control of movement through the button array.



Winship et al. (2023), Fig. 1. Separate monitor and directional controller



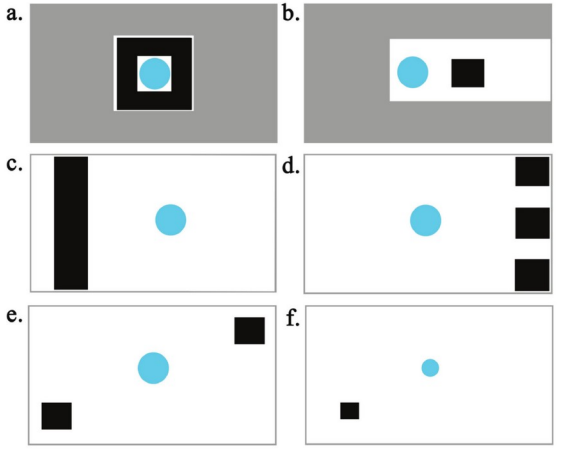
Meet_EVE_HD, 70–84 s. Sample and choice display.

Cursor contact determines the trial consequence

A **contingency** is a relationship between an event and the consequence that follows it. In the Cursor Training Game, cursor contact with a target produced a success tone and reinforcement. The animal's button presses controlled when that contact occurred.

The cursor was a bright blue circle against a white background, and the targets were black rectangles. Target location and cursor size changed across training phases. These changes altered the movements required for success while retaining the basic response–consequence relationship.

Game software recorded directional presses, their durations and response times. A successful response could therefore be evaluated from the cursor's contact with the target and from the sequence of inputs that produced it, rather than being scored only from a trainer's impression.



Winship et al. (2023), Fig. 2. Successive cursor-training task displays



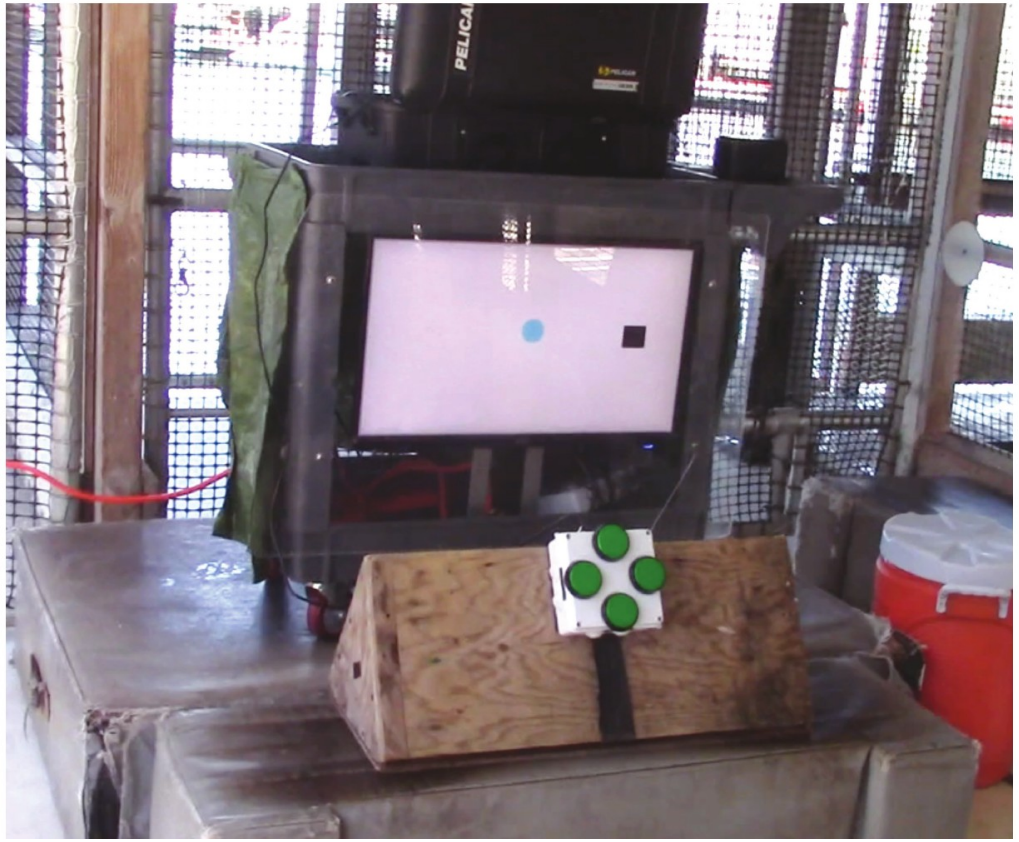
Meet_EVE_HD, 84–101 s. Response and trainer reward.

A success tone becomes a conditioned reinforcer

A **conditioned reinforcer** is a signal that acquires reinforcing value through its association with other consequences. The EVE success tone was paired with reinforcement during training. The animals subsequently responded to it in a manner resembling their familiar verbal success cue.

The tone could be delivered manually while the relationship between the apparatus and reward was established. Later, software generated it when the cursor contacted a target. The same signal therefore connected early training to the consequences of independent gameplay.

Food, ice and trainer encouragement were used as reinforcements. The tone marked the relevant event, while the additional consequence followed it. This temporal organization reduced reliance on a trainer's visible movement to identify which cursor action had been successful.



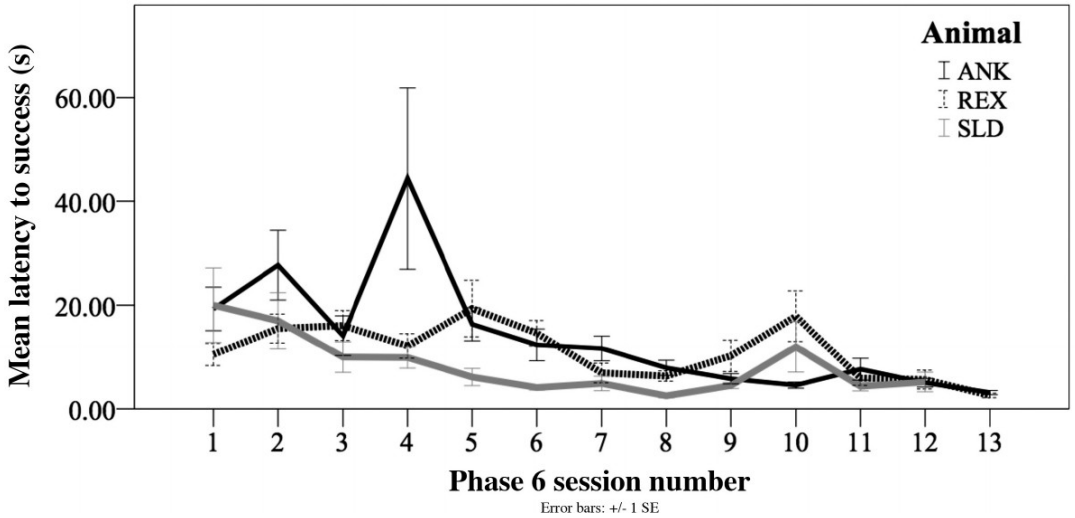
Winship et al. (2023), Fig. 1. Apparatus used for tone-conditioned gameplay

Target-contact latency declines during training

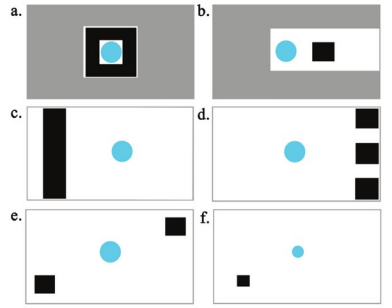
Latency is the time between a defined starting event and a response. EVE latency to success was measured from the first button contact to cursor contact with the target. This definition excludes the time before the animal initiated that trial's movement.

Across the final training sessions, latency to target contact declined for each sea lion. The result concerns improvement within a common single-target phase, after the earlier steps had established how to operate the directional controller.

Faster target contact can result from more direct routes, appropriately timed releases or reduced periods without effective input. The latency measure combines these possibilities. Recorded button sequences provide additional information about how the same final contact was achieved and how performance changed across sessions.



Winship et al. (2023), Fig. 4. Latency across final training sessions



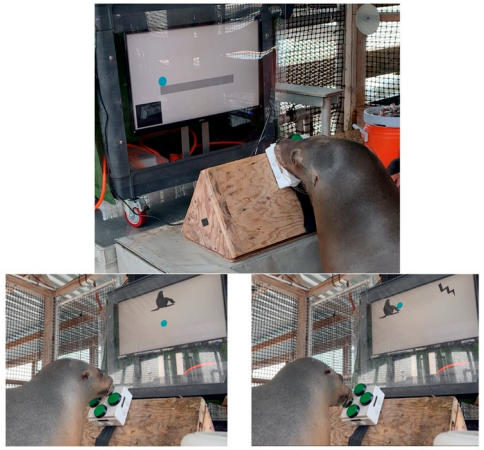
Winship et al. (2023), Fig. 2. Single-target task underlying the latency measure

Maze choices determine where the cursor falls

The MAZE game restricted cursor movement to left and right choices. The cursor traveled along platforms and could fall toward a target below them. A correct choice of direction and drop location allowed target contact; an incorrect route could send it off the display.

Planning concerns selecting actions in relation to a later outcome. The maze imposed such an opportunity by making the consequence of a horizontal choice depend on the platform below. Introductory trials simplified this relationship before more complex arrangements were presented.

Following advancement, all focal animals were offered the familiar cursor task again. Its target-contact requirement retained the directional responses acquired during training. Records from those sessions linked the animals' experience in the maze or matching games to their subsequent movement times and button use in the common cursor task.



Roberts et al. (2023), Fig. 3. Maze training arrangement and matching alternatives



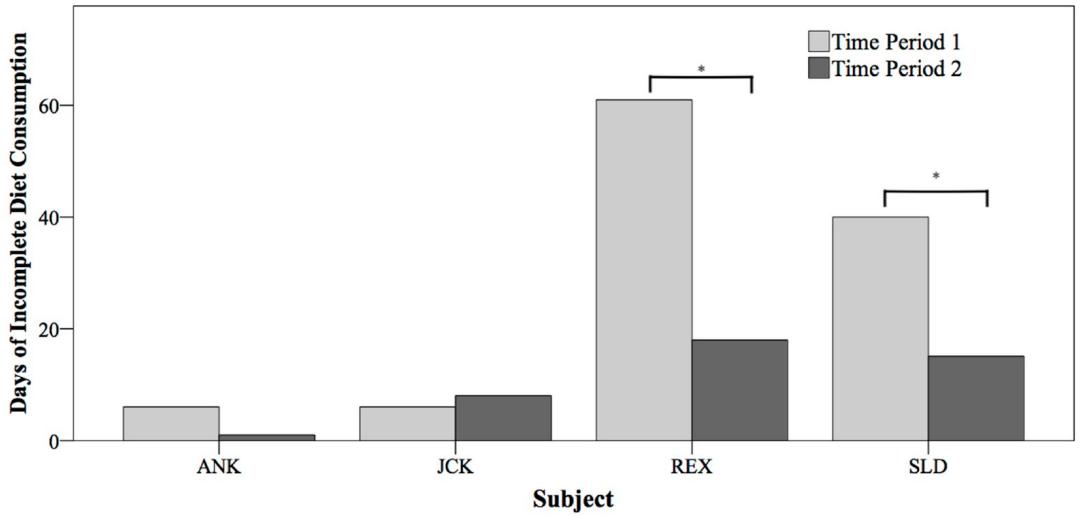
Meet_EVE_HD, 53–59 s. Maze gameplay.

Feeding records change after cognitive enrichment

Winship and colleagues compared feeding records from two-year periods before and after EVE was introduced. **Incomplete consumption of the offered diet** meant that the animal left part of its planned ration unfinished. The records supplied a daily behavioral outcome separate from the response time measured in a computer trial.

Incomplete-consumption days declined from 61 to 18 for REX and from 40 to 15 for SLD after EVE began. The analysis distinguished feeding records from days removed because of clinical illness. These counts described how often each animal left an unfinished ration within the analyzed periods.

The average percentage of the offered diet consumed also rose, from about 97% to 99% for REX and from about 98% to 99% for SLD. More complete daily intake and fewer unfinished-ration days described related changes in feeding. The study connected the enrichment period with repeated observations of participation and food consumption in managed animals.



Winship et al. (2024), Fig. 2. Incomplete-diet-consumption days across two periods



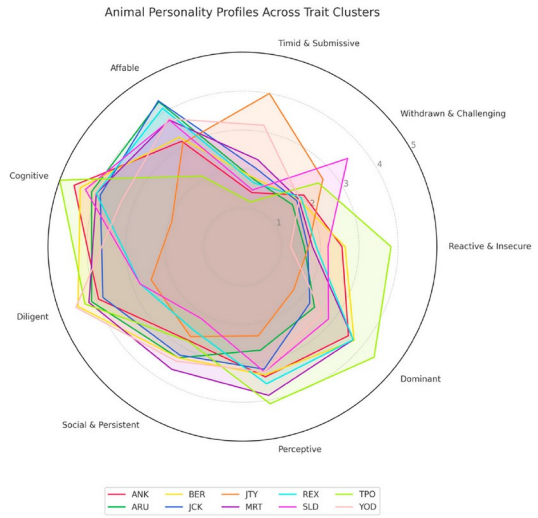
Winship et al. (2024), Fig. 1. EVE session context for the feeding records

Rated perceptiveness is associated with faster responses

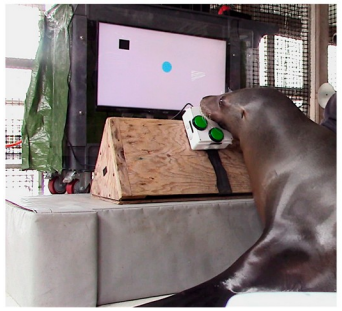
Ramos and Winship used ratings from familiar trainers to describe recurring behavioral tendencies. **Personality traits** are patterns in how an individual tends to behave across situations. The study related ratings such as Perceptiveness and Dominance to response-time and button-press measures during computerized tasks.

Higher Perceptiveness ratings were associated with faster target-contact responses in both initial and later sessions. Higher Dominance ratings were associated with faster initial responses and fewer initial button presses. Response latency and total input count described different aspects of the animal's performance.

An animal could differ in response speed, in the number of inputs used, or in both measures. The Perceptiveness association persisted across the training period. The study connected behavioral ratings with recorded performance in a defined cognitive task, while measuring the dimensions of that performance separately.



Ramos & Winship (2025), Fig. 5. Trait profiles related to response measures



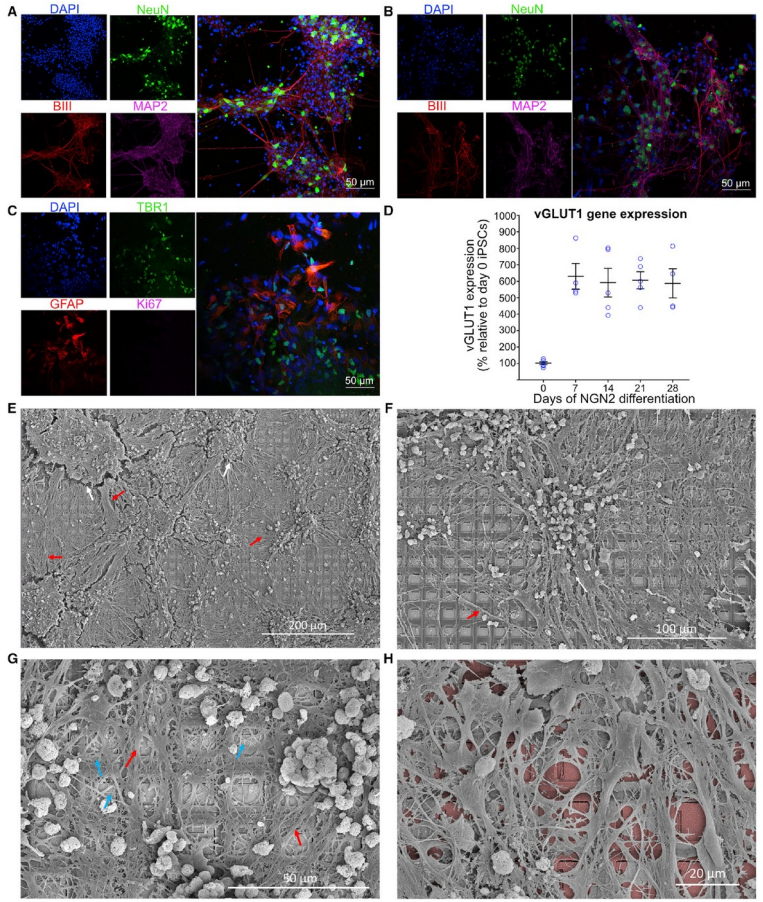
Ramos & Winship (2025), Fig. 1. Computerized task used for response measurements

Living cortical cultures form interconnected networks

Living neurons can form electrically active networks outside the body. Kagan and colleagues cultured cells from embryonic mouse cortex and from human induced pluripotent stem cells, then connected these preparations to an electronic interface. **Induced pluripotent stem cells** are reprogrammed cells that can develop into many specialized cell types.

The **cortex** is the layered outer region of the cerebral hemispheres. In these **dissociated cultures**, cells separated from the source tissue were dispersed onto a recording surface. Fluorescent labeling identified neuronal cell bodies and their processes. **Dendrites** receive inputs, while **axons** convey outgoing neural signals. **Astrocytes** are a type of non-neuronal glial cell that supports neuronal function.

The resulting networks provided the living component of DishBrain. Signals traveling through their processes could reach different recording sites, while external stimulation could influence subsequent activity. The study linked this neural activity to movement of a virtual Pong paddle, giving cultured networks an experimentally controlled relationship with an external task.



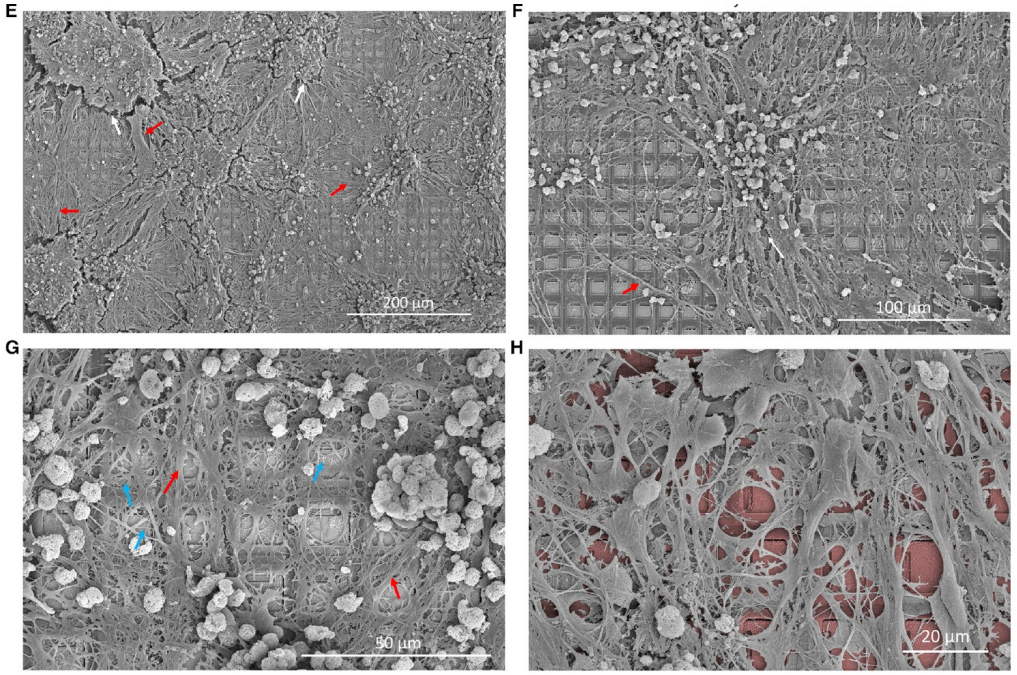
Kagan et al. (2022), Fig. 2(A–H). Neurons, neurites, astrocytes, and glutamatergic markers in cortical cultures.

Electrode arrays connect neural signals with electronics

A **microelectrode array**, or MEA, contains many small conducting sites that record voltages near cells and deliver electrical pulses. DishBrain used a high-density array on which neuronal processes extended across multiple electrodes.

An **action potential** is a rapid electrical event that propagates along a neuron. An extracellular electrode measures voltage changes outside the cell as this activity passes nearby. It therefore samples local activity without placing a recording pipette inside each neuron.

Scanning electron microscopy resolved dense cellular coverage over the array after prolonged culture. Dendritic and axonal connections spanned the recording surface rather than forming one neuron for each electrode. This geometry allows electronics to interact with distributed neural tissue: an input at one location can affect activity detected elsewhere through the living network.



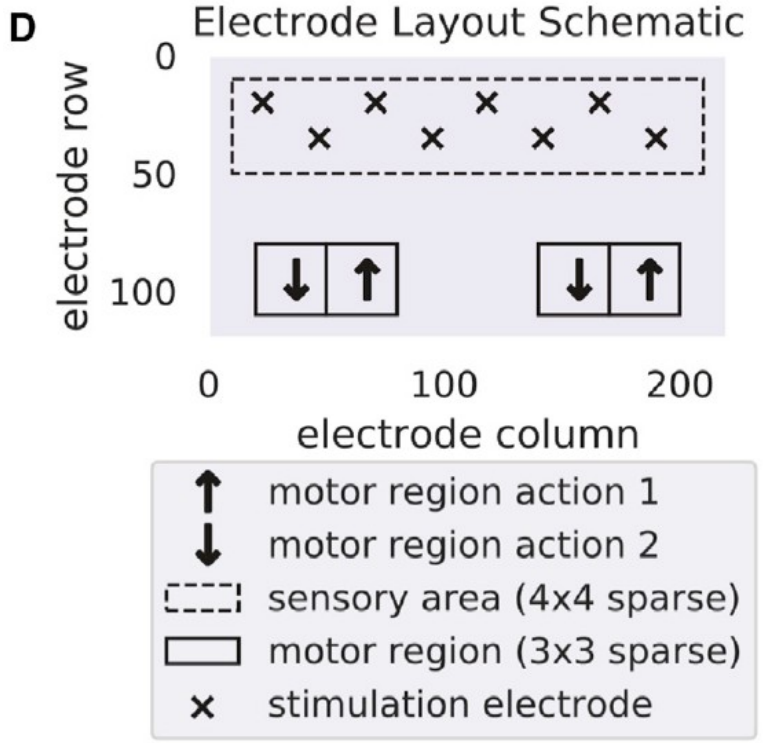
Kagan et al. (2022), Fig. 2(E–H). Cultured neurons and processes span the electrode surface.

Place and rate codes convey the virtual ball position

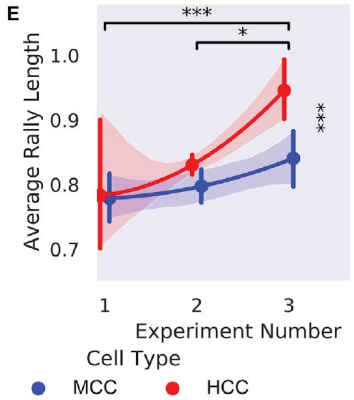
The virtual Pong ball was converted into stimulation delivered to eight predefined sensory electrodes. A **place code** represents a variable through the location of input; the active electrode corresponded to the ball's position along the game's vertical axis.

A **rate code** represents information through how frequently events occur. The refined protocol combined place coding with stimulation rates from 4–40 Hz to convey additional ball information. The researchers compared this richer input with earlier pilot configurations using less detailed sensory stimulation.

Gameplay performance improved across the refined encoding designs. The relationship connects the electronic representation of the environment with what the living network can use to guide an output. Information arrives as timed pulses at specific sites, making input design a functional part of the biological task rather than simply a display of a game on a screen.



Kagan et al. (2022), Fig. 4(D). Electrode positions assigned to sensory and motor functions.



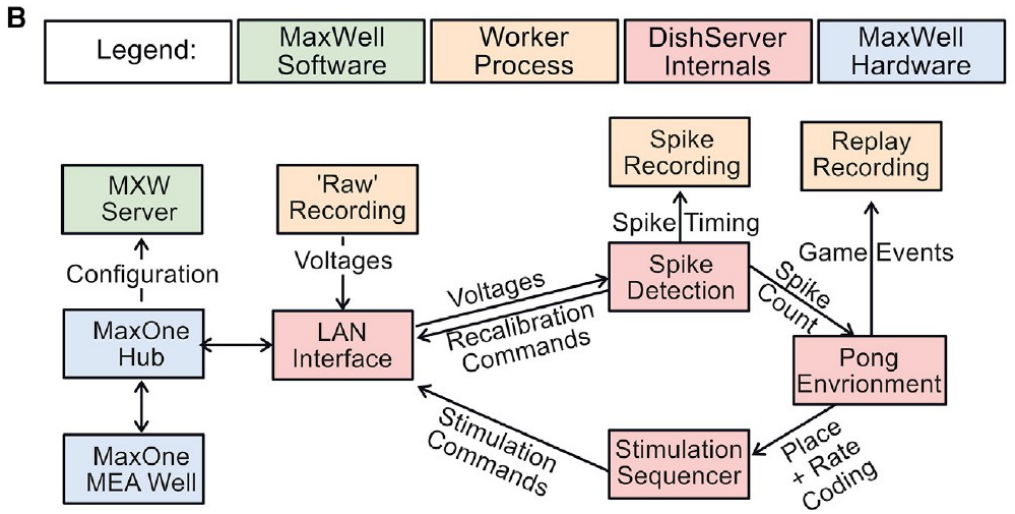
Kagan et al. (2022), Fig. 4(E). Gameplay performance changes across sensory-encoding designs.

A closed loop links neural output to subsequent input

A **closed-loop system** uses its measured output to determine later input. DishBrain recorded neural activity, used that activity to move a virtual paddle, and delivered electrical stimulation based on the resulting state of the game.

The software exchanged data with the electrode hardware at low latency. The reported packet-processing latency was typically below 50 ms. **Latency** is the delay between an event and the processing or response associated with it; short delays maintain the temporal relationship between output and its consequences.

The loop therefore creates a causal interaction between cultured tissue and the virtual environment. Activity changes the paddle, paddle movement changes whether the ball is intercepted, and that outcome changes stimulation. Repeated interaction gives the network input contingent on its own output rather than a sequence of stimuli that proceeds independently of neural activity.



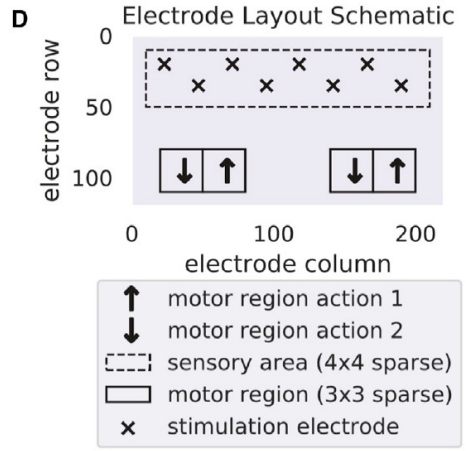
Kagan et al. (2022), Fig. 4(B). Published data paths connecting neural activity and the game.

Motor-region activity determines paddle movement

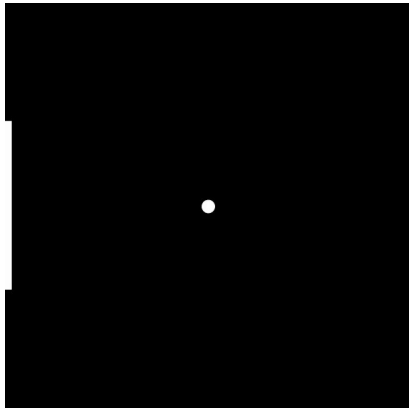
DishBrain assigned different groups of recording electrodes to two motor functions. **Motor region** here means a software-defined portion of the array whose recorded activity moves the paddle, rather than an anatomically identified motor cortex.

Activity in one region moved the paddle upward, while activity in the other moved it downward. The researchers tested configurations and used weighting to account for differences in regional activity. The final layout separated sensory inputs from the counterbalanced motor assignments used to evaluate output.

Successful control therefore depends on the balance and timing of regional activity rather than merely producing spikes somewhere in the culture. The assignment converts distributed neural signals into a measurable action. Because cells were not organized according to that software label before testing, the task probes how an existing living network changes its activity when that activity acquires external consequences.



Kagan et al. (2022), Fig. 4(D). Electrode positions assigned to sensory and motor functions.



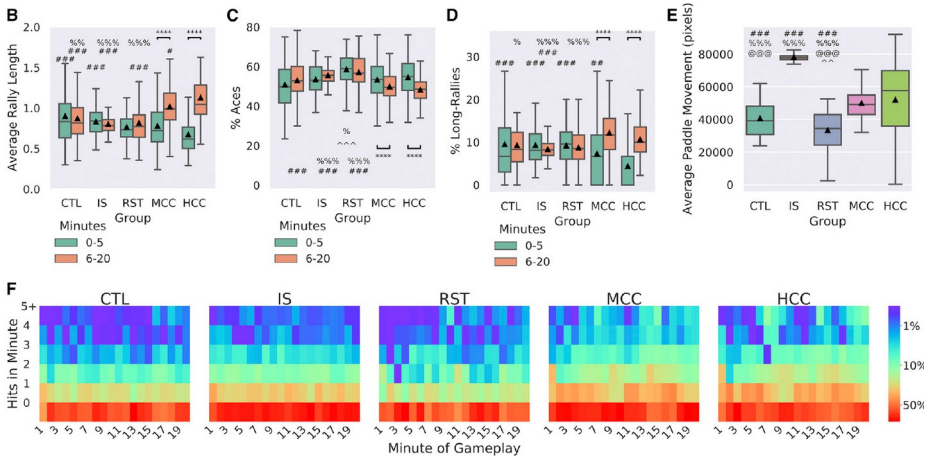
Kagan et al. (2022), Video S1. Neural control of a Pong paddle.

Mouse and human cultures improve during gameplay

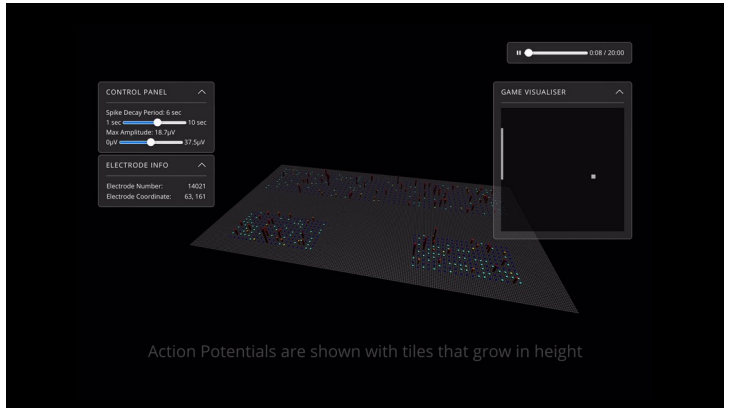
Average rally length measures how many consecutive successful interceptions occur before a miss. Kagan and colleagues compared performance during the first 0–5 min of a session with performance during minutes 6–20.

Both mouse and human cortical cultures increased rally length over time under the feedback-enabled task. They also reduced immediate misses, called **aces**, and increased the occurrence of longer rallies. These complementary measures connect successful interception with persistence across successive ball returns.

The resulting improvement depended on interactions during the task rather than simply on the presence of active cells. The same interface measured performance from biological and nonbiological controls. Multiple gameplay outcomes therefore supplied converging behavioral readouts of adaptation within this defined virtual environment, while retaining separate measures of what happened in the neural recordings.



Kagan et al. (2022), Fig. 5(B–F). Gameplay improvement, paddle movement, and hit distributions in experimental and control groups.



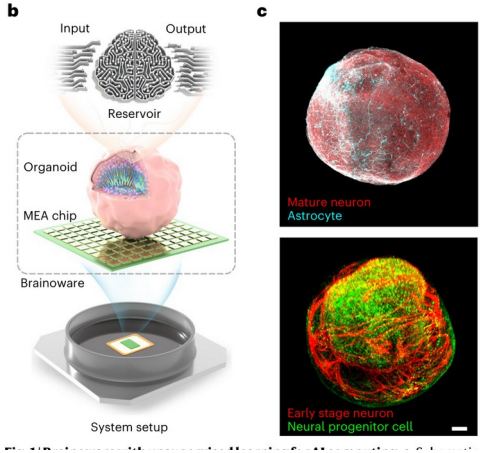
Kagan et al. (2022), Video S2. SpikeStream activity and Pong control.

An organoid transforms a changing electrical input

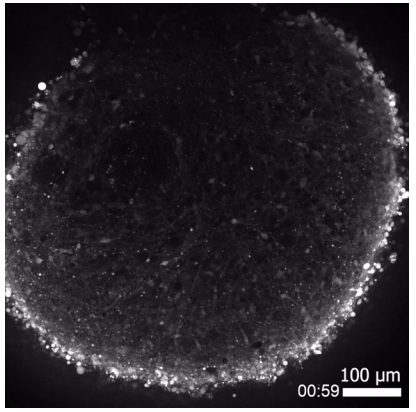
A **reservoir** transforms a time-varying input into a rich pattern of internal states that a simpler readout can use. Cai and colleagues mounted a human brain organoid on a high-density electrode array, using living neural activity as the physical reservoir in Brainoware.

A **brain organoid** is a three-dimensional tissue aggregate developed from pluripotent cells through **differentiation**, the development of specialized cellular features, and self-organization. Immunostaining identified neurons, neural progenitor cells, and astrocytes in the preparation. Electrical stimulation entered the tissue through the array, and evoked activity supplied the output features.

The electronic readout mapped these features onto a computational answer. Unlike the Pong paddle loop, this arrangement separated tissue responses from a trained classification or prediction function. The biological reservoir transformed input, while the readout supplied the task-specific mapping from measured neural activity to the desired output.



Cai et al. (2023), Fig. 1B–C. Organoid-electrode arrangement and neuronal tissue markers.



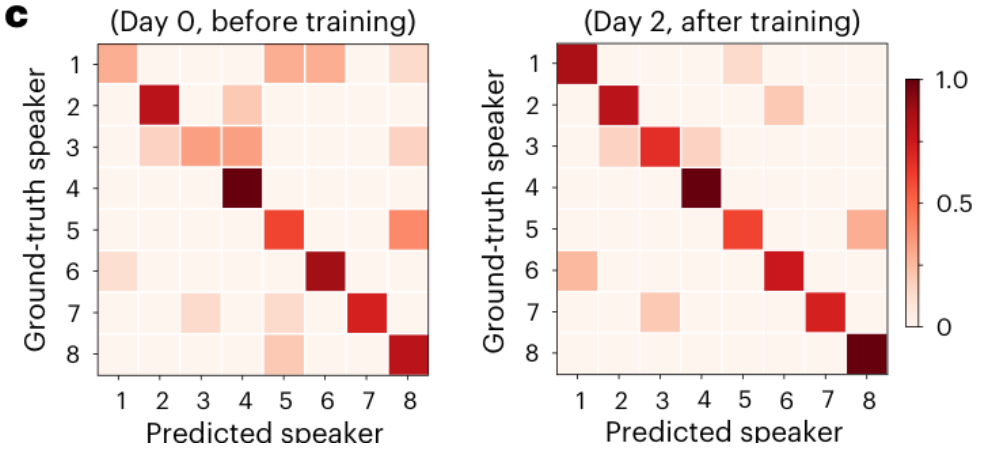
Cai et al. (2023), Supplementary Video 2. Confocal calcium activity (publisher 10× playback).

Speech classification uses a trained readout

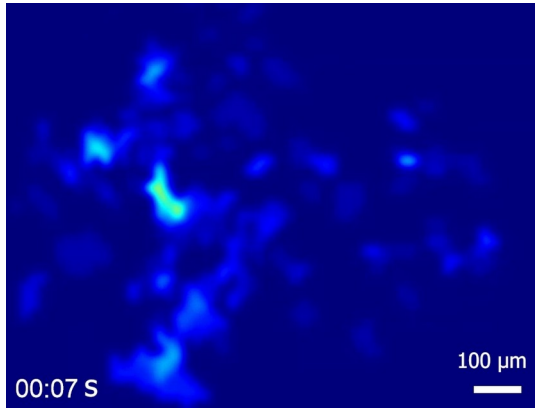
Brainoware classified speakers from isolated Japanese vowel recordings. The task used /a/ and /e/ utterances from eight male speakers, converted into spatiotemporal electrical stimulation. **Classification** assigns an input to a category, here the identity of the speaker who produced the recording.

Evoked organoid activity entered a logistic-regression readout, an algorithm trained to map response features onto speaker labels. Training also repeatedly exposed the organoid to the encoded recordings. The experiment therefore included supervised readout fitting alongside repeated stimulation of an adaptive biological reservoir.

Classification accuracy increased from about 51% before training to about 78% after four epochs over two days. An **epoch** is one presentation of the training dataset. The behavioral analog is recognition of a category from structured input, implemented jointly by the tissue's response transformation and the electronically trained output mapping.



Cai et al. (2023), Fig. 3C. Speaker classification before and after training.



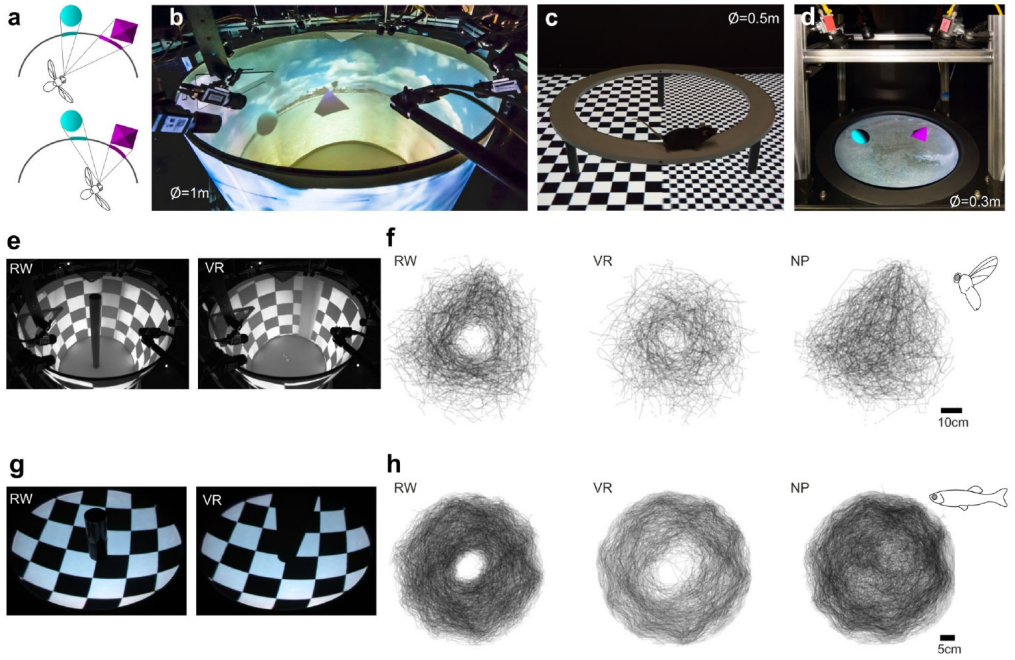
Cai et al. (2023), Supplementary Video 1. Organoid spike-activity heat map (publisher 4× playback).

Animal action changes the next visual input

Virtual reality in animal experiments is a controlled coupling between an animal's actions and its sensory input. A camera or treadmill sensor measures movement; a computer updates the visual environment in response. The animal's next movement therefore depends on a world that its previous movement helped determine.

Stowers and colleagues built **FreemoVR**, a system that tracks freely walking mice, flying fruit flies, and swimming zebrafish. Display surfaces surround the animals, while game-rendering software calculates the scene from their current location. Natural body movement continues as visual objects and backgrounds change under computer control.

Flies circled both a real post and its virtual counterpart. Without a post, they also occupied the arena center. Zebrafish similarly avoided the space occupied by real or virtual posts, connecting controlled visual geometry to the animals' trajectories.



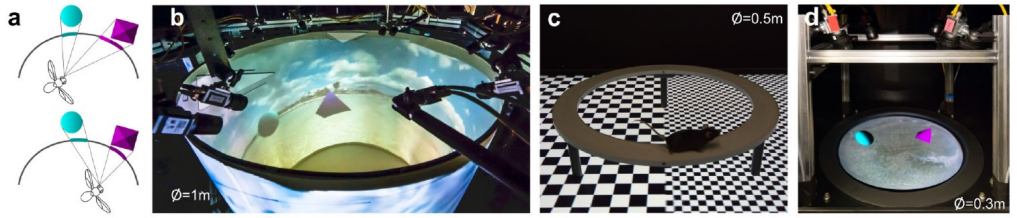
Stowers et al. (2017), Fig. 1(a–h). Free-moving arenas and responses to real and virtual posts.

Head tracking preserves the visual perspective

Perspective is the dependence of an object's projected appearance on the observer's location. A nearby post moves across the retina differently from a distant wall when an animal changes position. FreemoVR tracks this position and recalculates the display, preserving the corresponding relationship between movement and visual input.

Camera-based **computer vision**, the extraction of position from recorded images, supplies the location estimate. Spatial calibration links camera coordinates to display surfaces. For swimming fish, tracking also compensates for the bending of light at the air–water boundary, so the rendered view corresponds to the underwater observer.

The apparatus supported cylindrical flight arenas, projected mouse platforms, and aquatic arenas. Flies and fish changed their paths around virtual posts as they did around real posts. Rendering from the animal's viewpoint supplies object-related movement cues even though the display surface itself stays stationary.



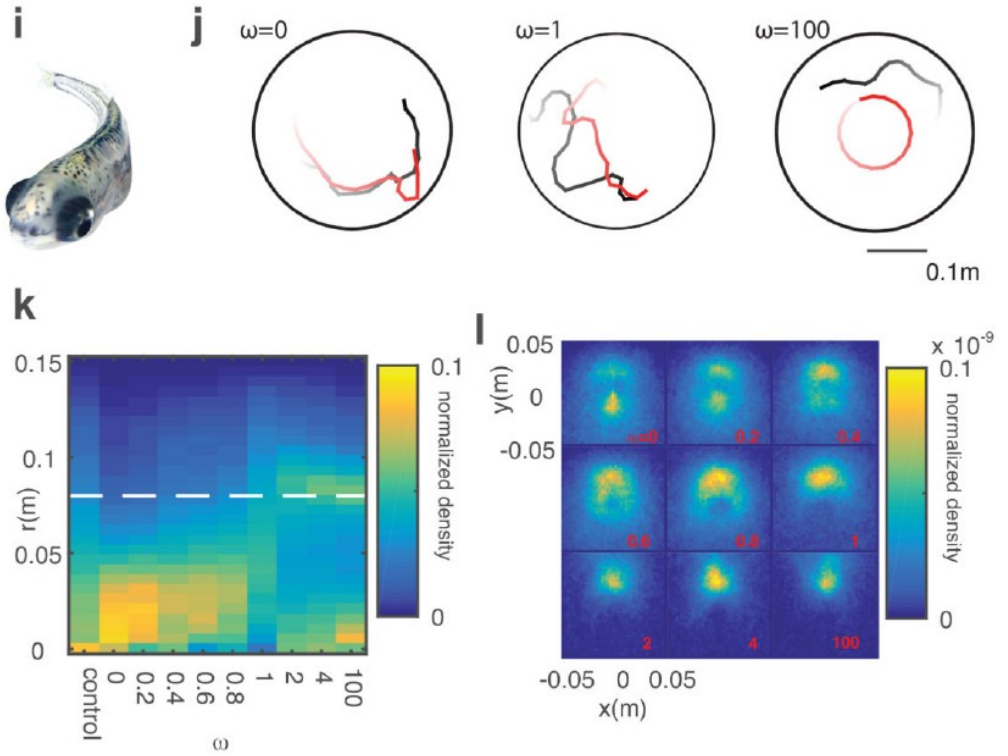
Stowers et al. (2017), Fig. 1(a–d). Perspective rendering and flight, mouse, and aquatic apparatus.

Virtual partners alter zebrafish social movement

Zebrafish interacted with a photorealistic virtual fish whose motion was controlled by the experimenter. A **virtual agent** is a computer-controlled entity that changes its behavior according to specified rules. Its response to the real animal can be varied independently of its appearance.

The study combined the virtual fish's preferred direction with its attraction toward the real fish. An agent driven almost entirely by its own directional preference tended to lose its follower. Strong attraction kept the pair together but reduced the agent's ability to impose its preferred direction.

Intermediate coupling balanced these competing influences and increased successful following. Occupancy maps and trajectories linked leadership to a reciprocal movement rule: a prospective leader maintained social contact while expressing its own directional tendency. Real fish therefore supplied feedback that changed the virtual partner's actions.



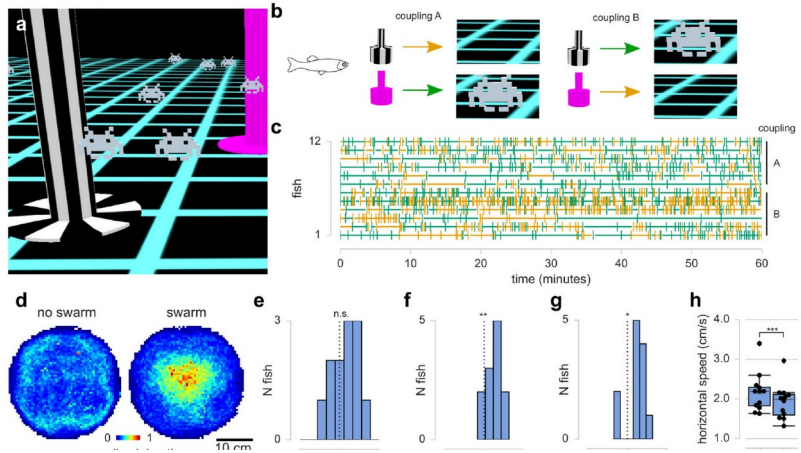
Stowers et al. (2017), Fig. 5(i–l). Real–virtual fish trajectories and effects of social coupling.

Portals change a fish’s world without moving its body

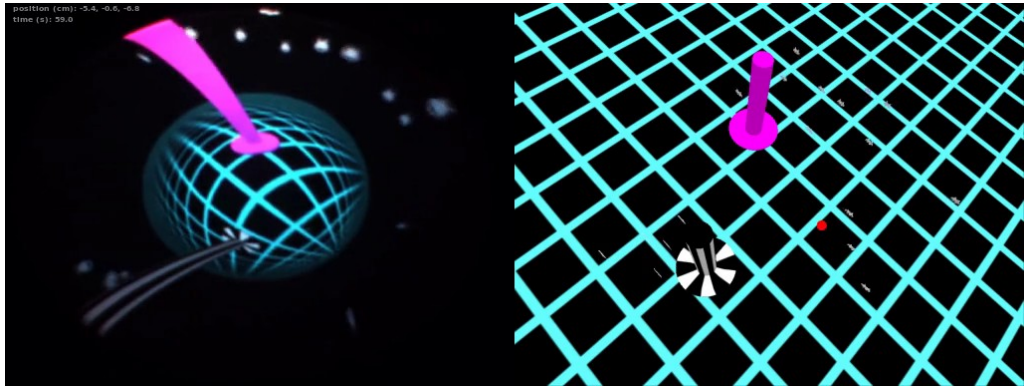
Zebrafish could enter visually marked **portals**, regions that triggered an immediate change of the displayed environment. The fish’s body stayed in the same tank while the visual world changed. Two portal appearances were coupled to different outcomes, separating the choice of entrance from behavior after the scene changed.

One environment contained a swarm of game-like agents. These agents followed a **flocking rule**, a movement algorithm that coordinated their motion with nearby individuals, including the real fish. Fish occupancy and swimming speed changed in the swarm condition; animals spent longer there and swam more slowly.

The fish preferred magenta portals, but their entrance choices did not favor the swarm outcome. Longer residence in a scene therefore differed from learning which entrance produced that scene. Portal color, entry choice, and post-entry movement represented distinct behavioral measurements in the same virtual task.



Stowers et al. (2017), Fig. 5(a–h). Portal choices, scene occupancy, and swimming with a virtual swarm.



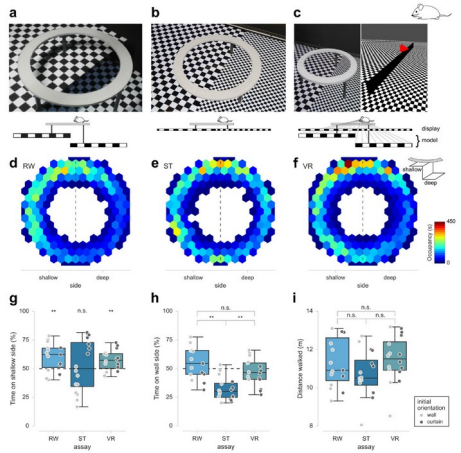
Stowers et al. (2017), Supplementary Video 13. Zebrafish portals and virtual swarms.

Freely moving mice avoid virtual depth

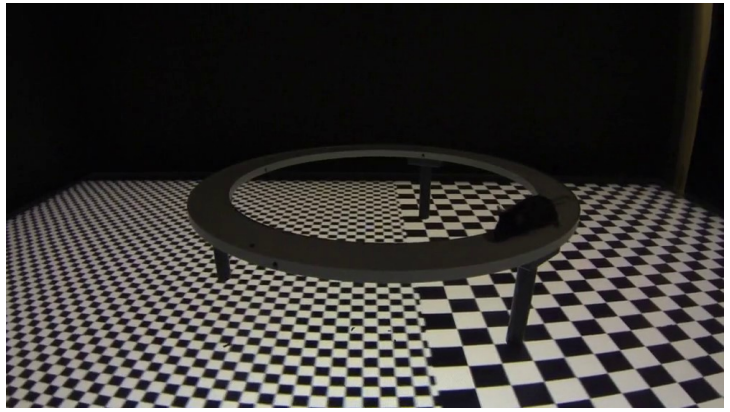
FreemoVR mice walked on a circular platform above checkerboard surfaces that specified shallow or deep space. The real condition placed surfaces 20 and 40 cm below the track. The virtual condition reproduced the corresponding views while tracking the mouse's changing position on the platform.

Motion parallax is the different apparent movement of objects at different distances during an observer's movement. Mice spent more time above the shallow region in both real and perspective-correct virtual conditions. Their physically identical platform supported different choices depending on the visually specified depth below it.

A static-display control retained depth-related texture appearances while removing correct movement-dependent parallax. That condition lacked the same shallow-side preference. The comparison linked avoidance to visual changes generated by self-motion, rather than to the mere presence of larger or smaller checkerboard elements.



Stowers et al. (2017), Fig. 2(a–i). Real, static, and perspective-correct depth conditions with occupancy measures.



Stowers et al. (2017), Supplementary Video 8. Mouse elevated-track depth assay.

Key takeaways

Biological explanation Generalization identifies recurring relationships in behavior; reduction explains the physiological processes that produce them.

Plasticity and development Experience and development change neural organization, including relationships among networks after early brain injury.

Evolution and ethics Inherited variation and natural selection shape behavior; human and animal research includes defined protections and institutional oversight.

Optical circuit control Light-sensitive proteins translate illumination into membrane currents, allowing targeted changes in neuronal activity and behavior.

Behavioral measurement Neural recordings and computer vision connect measured signals with the animal’s movements and environment.

Closed-loop learning A control mapping links an action to a changed input; feedback can alter future behavior in whole animals and living neural networks.

Carlson & Birkett (2022); Kliemann et al. (2019); Boyden et al. (2005); Mathis et al. (2018); Givon et al. (2022); Kagan et al. (2022)